Poisson Statistics for Values of Binary Quadratic Forms in Short Intervals

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Abstract

We study the number of integers represented by a binary quadratic form in short intervals. Extending the work of Freiberg, Kurlberg, and Rosenzweig on sums of two squares, we formulate an analogue of the Hardy–Littlewood prime k-tuple conjecture for the integers represented by a genus of primitive positive definite binary quadratic forms of a fixed negative discriminant. We show that this hypothesis implies that the number of integers represented by a genus, and by each class in the genus, in a random interval $(n, n+y]$ with $n \leq x$, where y is τ times the average spacing $\sqrt{\log x}/b(d)$, is asymptotically Poisson distributed with parameter τ. The proof uses the method of moments, with Bernays' asymptotic for the number of represented integers in place of the prime number theorem, and an adaptation of Ford's simplified argument for the average of the singular series. We also give numerical evidence for the hypothesis.

Contents

1 Introduction **2**
1.1 Notation and main results . . . 3
1.2 Outline of the proof and organization . . . 3
1.3 Acknowledgments . . . 4
1.4 Use of Large Language Models . . . 4

2 Two inputs from Bernays' dissertation **4**

3 Theorem 1.1 from Theorem 1.2 **5**

4 k-point correlations and the singular series **6**

5 Local densities at odd primes: a unified recursive treatment **7**
5.1 Setup: completing the square . . . 7
5.2 The nonsingular case: a single Hensel lemma . . . 8
5.3 The trichotomy mod p . . . 8
5.4 Split primes: resolved directly, no recursion . . . 9
5.5 Inert primes: a self-referential recursion . . . 9
5.6 Primes dividing d: the degenerate case, recursively . . . 9
5.7 Regularity, uniformly . . . 11
5.8 Multi-point densities: near 1 away from $\det \mathbf{h}$ . . . 13

6 Local densities at $p = 2$: a parallel recursive treatment **14**
6.1 The key classical input . . . 14
6.2 Setup: two completing-the-square regimes . . . 14
6.3 The case $2 \nmid d$: split and inert at 2 . . . 15
6.4 The case $4 \mid d$: four base types, then a recursion in the conductor . . . 16
6.5 Regularity at $p = 2$ . . . 17

7 The singular series and its average **18**
7.1 A Fubini/CRT averaging identity 18
7.2 Convergence of the singular series 19
7.3 The Ford-style average 19

8 Proof of Theorem 1.2 **23**

9 Numerical evidence for Hypothesis $U(k)$ **24**
9.1 Method 24
9.2 2-point and 3-point correlations 24
9.3 Averages and the dependence on x 27
9.4 The distribution in short windows 27

1 Introduction

By the prime number theorem, the number of primes up to x is asymptotic to $x/\log x$, so that an interval $(n, n+y]$ with $n \le x$ and $y = \lambda \log x$ contains λ primes on average. In 1976 Gallagher [Gal76] showed that, assuming a uniform version of the Hardy–Littlewood prime k-tuple conjecture, the number of primes in such an interval is asymptotically Poisson distributed with parameter λ, as n ranges over the integers up to x. In other words, at the scale of their average spacing the primes behave like independent random points of density $1/\log x$. Gallagher's proof is by the method of moments: the k-th moment of the number of primes in the interval is a sum over k-tuples of shifts, the hypothesis replaces the count of each prime k-tuple by its singular series, and the heart of the argument is an estimate showing that the singular series equals 1 on average over k-tuples of shifts in a box. Ford [For16] later found a much simpler proof of this estimate.

For the sums of two squares, the role of the prime number theorem is played by a theorem of Landau from 1908: the number of integers up to x that are sums of two squares is asymptotic to $Kx/\sqrt{\log x}$, where $K = 0.7642\ldots$ is the Landau–Ramanujan constant. These integers thus have average spacing about $\sqrt{\log x}/K$ near x, and the natural windows are $(n, n+y]$ with $y = \lambda\sqrt{\log x}/K$. Freiberg, Kurlberg, and Rosenzweig [FKR17], motivated by the level spacings of toral point scatterers, proved the analogue of Gallagher's theorem in this setting, assuming an analogous Hardy–Littlewood-type hypothesis for sums of two squares. Their work built on earlier results on the pair correlation of sums of two squares: Connors and Keating [CK97] conjectured a formula for it and found numerical evidence for Poisson spacings, and Smilansky [Smi13] expressed the singular series for pairs as a product of p-adic densities and showed that, assuming (a reformulation of) their conjecture, the second moment of the number of sums of two squares in short intervals is consistent with Poisson statistics. Smilansky also went beyond sums of two squares: for each of the forms $x^2 + dy^2$ with $d = 1, 2, 3, 4, 7$, which have class number one, Smilansky formulated an analogous pair correlation conjecture and showed that, assuming it, the second moment is again consistent with Poisson statistics [Smi13, §6]. These results involve only pair correlation and the second moment, whereas the theorem of Freiberg, Kurlberg, and Rosenzweig establishes convergence of all moments. As in Gallagher's work, the analytic heart of their argument is an estimate for the average of a singular series, and they point out that Ford's simplification applies there as well.

In this article we extend the theorem of Freiberg, Kurlberg, and Rosenzweig from sums of two squares to the integers represented by an arbitrary primitive positive definite binary quadratic form, or more precisely by a class or a genus of such forms of a fixed discriminant $d < 0$. Here the role of the prime number theorem is played by the dissertation of Bernays [Ber12] from 1912, written under Landau, which extends Landau's theorem to all such forms: for each class C of discriminant d, the number $N_C(x)$ of integers up to x represented by C satisfies:

$$N_C(x) \sim b(d)\,\frac{x}{\sqrt{\log x}} \qquad (x \to \infty),$$

for a constant $b(d) > 0$ that depends only on d, and not on C. The precise statement, together with a second result of Bernays comparing a class with its genus, is given in §2. The integers represented by C therefore have average spacing about $\sqrt{\log x}/b(d)$ near x, and we take windows of exactly this length: for fixed $\tau > 0$ we choose y so that $y\,N_C(x)/x \to \tau$, that is, $y \sim \tau\sqrt{\log x}/b(d)$, so that a window $(n, n+y]$ with $n \le x$ contains τ represented integers on average in the limit. We will show that, conditionally on a

uniform Hardy–Littlewood-type hypothesis, the number of represented integers in such a window is Poisson distributed with parameter τ. In particular, for the five forms considered by Smilansky [Smi13, §6], this passes from the second moment to the full distribution, under a hypothesis on correlations of every order.

1.1 Notation and main results

Throughout, $d < 0$ is a discriminant (not assumed fundamental), $C(d)$ denotes the form class group of order $h(d)$, and there are $g(d)$ genera, each containing $h(d)/g(d)$ classes; see [Cox13, §3.B] and [Cas78]. For a positive definite form f of discriminant d we write $E_f := f(\mathbb{Z}^2)$ for the set of integers it represents. This depends only on the class C of f, so we write $E_C := E_f$, and for a genus R we set $E_R := \bigcup_{C \subset R} E_C$. Their counting functions are:

$$N_C(x) := \#(E_C \cap [1, x]), \qquad N_R(x) := \#(E_R \cap [1, x]).$$

Further, p denotes a rational prime, v_p the p-adic valuation, $\mathbb{Z}_p$ the p-adic integers, μ_p normalized Haar measure on $\mathbb{Z}_p$ ($\mu_p(\mathbb{Z}_p) = 1$), and $\chi_d(p) = \left(\frac{d}{p}\right)$ the Kronecker symbol, so that for $p \nmid d$, p is split if $\chi_d(p) = 1$ and inert if $\chi_d(p) = -1$.

Our main results are the following, in which Hypothesis $U(k)$ is stated precisely in §4.

Theorem 1.1 (Class-level Poisson distribution, conditional on Hypothesis $U(k)$)**.** *Fix a discriminant $d < 0$, a class $C \in C(d)$, and $\tau > 0$, and assume that Hypothesis $U(k)$ holds for every $k \geq 1$ for the genus containing C. Let x and y be real parameters tending to infinity in such a way that $y\, N_C(x)/x \to \tau$. Then for every fixed integer $m \geq 0$:*

$$\frac{1}{x}\,\#\big\{1 \leq n \leq x :\ N_C(n+y) - N_C(n) = m\big\} \ \longrightarrow\ e^{-\tau}\,\frac{\tau^m}{m!} \qquad (x \to \infty).$$

Theorem 1.2 (Genus-level Poisson distribution, conditional on Hypothesis $U(k)$)**.** *Fix a discriminant $d < 0$, a genus R of discriminant d, and $\tau > 0$, and assume that Hypothesis $U(k)$ holds for every $k \geq 1$ for R. Let x and y be real parameters tending to infinity in such a way that $y\, N_R(x)/x \to \tau$. Then for every fixed integer $m \geq 0$:*

$$\frac{1}{x}\,\#\big\{1 \leq n \leq x :\ N_R(n+y) - N_R(n) = m\big\} \ \longrightarrow\ e^{-\tau}\,\frac{\tau^m}{m!} \qquad (x \to \infty).$$

Since $N_R(x) = N_C(x)(1+o(1))$ for every class $C \subset R$ (§2, Inputs A and B), the conditions $y\, N_C(x)/x \to \tau$ and $y\, N_R(x)/x \to \tau$ are equivalent, so Theorems 1.1 and 1.2 apply to the same windows and no separate calibration is needed. For $d = -4$ there is a single class, that of $x^2 + y^2$, and $b(-4) = K$; in this case Theorem 1.1 recovers part (b) of [FKR17, Theorem 1.2], on the number of values in a window, but not part (a), on nearest-neighbor spacings. Our Hypothesis $U(k)$ is also stronger than the one assumed there: Freiberg, Kurlberg, and Rosenzweig only ask that the analogous estimate hold on average over dilates of bounded convex sets, whereas we ask that it hold uniformly. We do this mainly to simplify the argument, and an averaged hypothesis should also suffice, since the proof of Theorem 8.1 only uses Hypothesis $U(k)$ after summing over the shifts.

1.2 Outline of the proof and organization

The idea is to split the work into an analytic part and an arithmetic reduction. Essentially all of the analytic work is concentrated in Theorem 1.2: local densities, a singular series, and a moment computation. Theorem 1.1 is then deduced from it in §3, via a classical equidistribution theorem of Bernays [Ber12] showing that a class and its genus represent the same integers with an error negligible at the scale the Poisson statistics live at. We prove this reduction first, so that the rest of the article can focus exclusively on Theorem 1.2.

For the average of the singular series in §7 we do not use the more involved arguments of Gallagher [Gal76] and of Freiberg, Kurlberg, and Rosenzweig [FKR17]. Instead we adapt Ford's simplified approach [For16] to binary quadratic forms: we truncate the Euler product of the singular series at a small multiple of

$\log H$, replace the remaining finite product by a periodic function, and average it over complete periods using the Chinese remainder theorem. The price is a worse error term, a relative saving of only $1/\log\log H$ (Theorem 7.3), but this is more than enough for the method of moments.

The article is organized as follows. In §2 we state the two inputs from Bernays' dissertation, which we use without proof, and in §3 we deduce Theorem 1.1 from Theorem 1.2. In §4 we state Hypothesis $U(k)$. We compute the local densities in §5 and §6, study the singular series and its average in §7, and prove Theorem 1.2 by the method of moments in §8. Finally, §9 contains numerical evidence for Hypothesis $U(k)$.

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1.4 Use of Large Language Models

Claude Sonnet 5.5 was used to transcribe and format this article from a set of handwritten notes and was also used to help with the numerical analysis in §9. Claude Opus 5.5 was used to review the article. Claude Opus 4.8 was used to compute the values of $\delta_{\{0\}}(2)$ in §6. Its derivation relied on results from class field theory, with which the author is not familiar, so it is not included here. The derivation of $\delta_{\{0\}}(2)$ presented in this article, and all other proofs, were produced without the use of large language models.

2 Two inputs from Bernays' dissertation

The reduction of Theorem 1.1 to Theorem 1.2 requires two classical facts due to P. Bernays [Ber12]. Their proofs are omitted as they are beyond the scope of the present article; a detailed treatment can be found in [Ber12].

Input A (Per-class asymptotic)**.** *For any class C of discriminant d,*

$$N_C(x) \;\sim\; b(d)\,\frac{x}{\sqrt{\log x}} \qquad (x\to\infty),$$

for a constant $b(d)>0$ depending only on d — the same constant for every class of that discriminant, regardless of which genus it lies in.

Input B (Class-genus equidistribution)**.** *Let $h=h(d)$ denote the class number. There is an exponent $\theta=\theta(d)>0$ such that for a class C inside a genus R, the difference $\Delta := E_R\setminus E_C$ (which is also the symmetric difference, since $E_C\subseteq E_R$) satisfies*

$$\#(\Delta\cap[1,x]) \;=\; O_d\!\left(\frac{x}{(\log x)^{1/2+\theta}}\right).$$

Any θ with $0<\theta<1/h$ and $\theta\le 1/4$ is admissible. If $h\le 2$, then every class is its own genus and $\Delta=\emptyset$.

Input A is the asymptotic for a single class together with the equal-density statement of [Ber12, p. 92]. Input B is not stated in exactly this form in [Ber12]; it follows from [Ber12, p. 102] (an integer represented by a genus but not by every class in it is an *exceptional number* for some fundamental class) and [Ber12, §3, p. 117] (for each fundamental class, the exceptional numbers up to x number $O\big(x/(\sqrt{\log x}\,(\log x)^{\vartheta'/h})\big)$ for every ϑ' with $0\le\vartheta'<1$ and $\vartheta'\le h/4$), by taking $\theta=\vartheta'/h$.

Combining the two inputs, we obtain essentially for free that for any genus R of discriminant d and any class $C\subset R$, $N_R(x)=N_C(x)(1+o(1))\sim b(d)\,x/\sqrt{\log x}$. Thus the identical constant $b(d)$ governs every genus and every class of the fixed discriminant d at once, which is exactly what makes a separate window calibration unnecessary for Theorem 1.1 versus Theorem 1.2 (§3). Note that Input B says that a class and its genus differ by a set so sparse that it is invisible at the scale $\sqrt{\log x}$ that Input A operates at, indeed by a positive power of $\log x$ (the exponent θ of Input B).

3 Theorem 1.1 from Theorem 1.2

In this section we deduce Theorem 1.1 from Theorem 1.2, using the two inputs of §2. The idea is to couple the window counts of E_C and E_R: the two sets differ only by a very sparse set, so their window counts coincide exactly on an event whose probability tends to 1.

Proof (of Theorem 1.1 from Theorem 1.2). Fix the genus R and a class $C \subset R$, and fix $\tau > 0$. By Input A (combined with Input B, as shown in §2), $\rho_1(x) := N_R(x)/x \to 0$ like $b(d)/\sqrt{\log x}$, and $N_R(x) = N_C(x)(1 + o(1))$. Hence the hypothesis $y\, N_C(x)/x \to \tau$ of Theorem 1.1 is equivalent to:

$$y \cdot \rho_1(x) \to \tau \qquad (\text{so } y \sim \tau\sqrt{\log x}/b(d)).$$

This matches the calibration used for Theorem 1.2 in §8.

Let $\Delta := E_R \setminus E_C$. Then $E_C \subseteq E_R$, since a class is contained in its genus, and Δ is exactly the set on which the two counts could differ. For n drawn uniformly from the integers $1 \le n \le x$, let Y_x^R and Y_x^C denote the counts of E_R, respectively E_C, in the window $(n, n+y]$. We split the proof into two claims.

Claim 1: We have $Y_x^C = Y_x^R$ *exactly*, on the event that the window $(n, n+y]$ contains no point of Δ at all.

Proof of Claim 1: Since $E_C \subseteq E_R$ and the two sets differ only by Δ, we get:

$$Y_x^R - Y_x^C \;=\; \#\big(\Delta \cap (n, n+y]\big) \;\ge 0.$$

Hence the two counts agree precisely when $\Delta \cap (n, n+y] = \emptyset$. This proves Claim 1.

Claim 2: The event in Claim 1 has probability $\to 1$ as $x \to \infty$.

Proof of Claim 2: By Input B, we have $\#(\Delta \cap [1, 2x]) = O_d\big(x/(\log x)^{1/2+\theta}\big)$ for some fixed $\theta = \theta(d) > 0$; this is a safe covering range, since the window $(n, n+y]$ reaches at most $x + y < 2x$ for x large, because $y = o(x)$. Only the positivity of θ will be needed.

Now average over the random starting point $n \in (0, x]$. Writing $\#(\Delta \cap (n, n+y]) = \sum_{m\in\Delta} \mathbf{1}[n < m \le n+y]$ and swapping the order of summation, we find:

$$\sum_{n=1}^{x} \#(\Delta \cap (n, n+y]) \;=\; \sum_{m\in\Delta} \#\{n \in [1, x] :\; m - y \le n < m\}.$$

For each fixed m, the inner count is at most y regardless of m: it counts integers in $[m-y, m-1] \cap [1, x]$, and since m is an integer, the full interval $[m-y, m-1]$ contains exactly $\lfloor y \rfloor \le y$ integers. Moreover, the inner count is zero once $m > x + y$. So the right-hand side is at most $y \cdot \#(\Delta \cap [1, x+y]) \le y \cdot \#(\Delta \cap [1, 2x])$. Since n is uniform on the $\lfloor x \rfloor$ integers $1 \le n \le x$, dividing by $\lfloor x \rfloor \ge x/2$ we obtain:

$$\begin{aligned}
\mathbb{E}\big[\#(\Delta \cap (n, n+y])\big] \;&\le\; \frac{y}{\lfloor x \rfloor} \cdot \#(\Delta \cap [1, 2x]) \;\le\; \frac{2y}{x} \cdot \#(\Delta \cap [1, 2x]) \\
&=\; O_d\!\left(\frac{y}{(\log x)^{1/2+\theta}}\right) \overset{(*)}{=}\; O_d\!\left(\frac{1}{(\log x)^{\theta}}\right) \;\longrightarrow\; 0,
\end{aligned}$$

where $(*)$ holds because $y = O(\sqrt{\log x})$. By Markov's inequality it follows that $\mathbb{P}\big(\#(\Delta \cap (n, n+y]) \ge 1\big) \to 0$ as $x \to \infty$. This proves Claim 2.

We now put our two claims to work. By Claim 2, with probability tending to 1 the window contains no point of Δ, so by Claim 1 we have $Y_x^C = Y_x^R$ exactly on an event of probability $\to 1$. Since $Y_x^R \to \text{Poisson}(\tau)$ in distribution by Theorem 1.2, the same must be true of Y_x^C. Since $Y_x^C = N_C(n+y) - N_C(n)$ takes integer values, this means precisely that $\mathbb{P}(Y_x^C = m) = \frac{1}{\lfloor x \rfloor}\#\{1 \le n \le x :\; N_C(n+y) - N_C(n) = m\} \to e^{-\tau}\tau^m/m!$ for every fixed $m \ge 0$. Since $\lfloor x \rfloor / x \to 1$, the same limit holds with $\lfloor x \rfloor$ replaced by x. This completes the proof of Theorem 1.1, granted Theorem 1.2. □

Remark 3.1. This coupling argument is why no separate class level hypothesis is necessary. To see why the coupling is needed, consider the naive alternative of running the method of moments for Y_x^C directly, comparing it with Y_x^R tuple by tuple. Write $\rho_k^C(\mathbf{h}; x)$ and $\rho_k^R(\mathbf{h}; x)$ for the k-point correlations (Definition 4.1)

of E_C and E_R. Since $E_C \subseteq E_R$, an integer n counted by ρ_k^R but not by ρ_k^C must have $n + h_i \in \Delta$ for some i, so the only bound that Input B provides is:

$$0 \;\le\; \rho_k^R(\mathbf{h}; x) - \rho_k^C(\mathbf{h}; x) \;\le\; \frac{1}{x} \sum_{i=1}^{k} \#\{1 \le n \le x : \; n + h_i \in \Delta\} \;\le\; \frac{k}{x}\, \#(\Delta \cap [1, 2x]) \;\ll_{k,d}\; \frac{1}{(\log x)^{1/2+\theta}}.$$

The k-th factorial moment is a sum of k-point correlations over the $\asymp y^k$ tuples of shifts in the window, so summing this bound over all of them gives a total error of size:

$$\frac{y^k}{(\log x)^{1/2+\theta}} \;\asymp_{k,d,\tau}\; (\log x)^{(k-1)/2-\theta},$$

to be compared with a main term of size τ^k. For $k = 1$ this error is $(\log x)^{-\theta} \to 0$, which is exactly the estimate used in the proof of Claim 2. For every $k \ge 2$, however, the exponent is $(k-1)/2 - \theta \ge 1/2 - \theta > 0$, since $\theta \le 1/4$ by Input B, so the error bound tends to infinity and the naive comparison fails already for the second moment, whatever the value of τ. To make it work one would need to know how Δ correlates with E_R, that is, a version of Hypothesis $U(k)$ for each class separately. The coupling argument avoids this altogether: it only uses the first moment of $\#(\Delta \cap (n, n+y])$, and since the event $Y_x^C = Y_x^R$ has probability tending to 1, the full Poisson distribution transfers for free.

With this reduction in hand, we turn to Theorem 1.2. Henceforth we will always drop the subscript R and write $E := E_R$, $N(x) := N_R(x)$, and so on, for the fixed genus under discussion.

4 k-point correlations and the singular series

In this section we state Hypothesis $U(k)$, the input on which Theorem 1.2 is conditional. It asserts that the k-point correlations of E are governed by a singular series built from local densities, in the spirit of the Hardy–Littlewood prime k-tuple conjecture. Throughout this section, unless otherwise stated, R is a fixed genus of discriminant $d < 0$, and $E = E_R$ and $N(x) = N_R(x)$ as at the end of §3.

To measure how likely it is that several integers $n + h_1, \dots, n + h_k$ are simultaneously represented, we count the corresponding values of n. This motivates the following definition.

Definition 4.1 (k-point correlation)**.** For $k \ge 1$ and a finite set $\mathbf{h} = \{h_1, \dots, h_k\} \subset \mathbb{Z}$ of distinct integers, the k-point correlation is:

$$\rho_k(\mathbf{h}; x) := \frac{1}{x} \sum_{1 \le n \le x} \mathbf{1}_E(n + h_1) \cdots \mathbf{1}_E(n + h_k), \qquad \rho_1(x) := \rho_1(\{0\}; x) = \frac{N(x)}{x}.$$

Remark 4.2. Our terminology follows [FKR17], where the analogous quantity $R_k(\mathbf{h}; x)$ is called the *correlation function*. This terminology comes from statistical mechanics and the study of spectral statistics, which is the setting of [FKR17]: there the sums of two squares arise as the energy levels of the square billiard.

For a prime p, let $S_p := S_p^{(R)} \subset \mathbb{Z}_p$ denote the set of values represented, over $\mathbb{Z}_p$, by (any form in) R; this is well defined since forms in the same genus are $\mathbb{Z}_p$-equivalent at every p. As the continuous image of the compact set $\mathbb{Z}_p^2$ under a polynomial map, S_p is compact, hence closed and Haar-measurable. This motivates the following definition.

Definition 4.3 (Local density)**.** For a prime p and a finite set $\mathbf{h} \subset \mathbb{Z}$ of distinct integers, the local density is:

$$\delta_{\mathbf{h}}(p) := \mu_p\Big(\bigcap_i (S_p - h_i) \Big).$$

For $\mathbf{h} = \{0\}$ this is $\mu_p(S_p)$. For instance, if $d = -3$ and R is the genus of $x^2 + xy + y^2$, then S_3 contains no element congruent to 2 modulo 3, so the translates S_3, $S_3 - 1$, $S_3 - 2$ have empty intersection and $\delta_{\{0,1,2\}}(3) = 0$. We will compute S_p exactly, at every prime, in §5–§6; knowing S_p explicitly then immediately shows how a truncated approximation to $\delta_{\mathbf{h}}(p)$ converges to it.

In §7 we will average the singular series over $\mathbf{h}$ by approximating it with a finite Euler product, truncated at primes $p \le Y$, whose factors are periodic in $\mathbf{h}$. For this we need a finite-level version of the local densities. This motivates the following definition.

Definition 4.4 (Truncated density)**.** For $\alpha \geq 1$ and a tuple $\mathbf{h} = (h_1, \ldots, h_k) \in \mathbb{Z}^k$, whose entries need not be distinct, the truncated density is:

$$\delta_{\mathbf{h}}^{(\alpha)}(p) := p^{-\alpha} \#\big\{0 \leq a < p^\alpha : \ a + h_i \in S_p \bmod p^\alpha \ \ \forall i\big\}.$$

Thus $\delta_{\mathbf{h}}^{(\alpha)}(p)$ asks that the reduction of $a + h_i$ modulo p^α lie in the reduction of S_p modulo p^α, for each i separately. Repeated entries are allowed because in §7 we average over all of $(\mathbb{Z}/p^\alpha\mathbb{Z})^k$; when the entries are distinct we identify $\mathbf{h}$ with the underlying set, which is harmless since the definition does not depend on their order. We now assemble the local densities into the singular series.

Definition 4.5 (Singular series)**.** For a finite set $\mathbf{h} \subset \mathbb{Z}$ of k distinct integers, the singular series is:

$$\mathfrak{S}_{\mathbf{h}} := \prod_p \lambda_{\mathbf{h}}(p), \qquad \lambda_{\mathbf{h}}(p) := \frac{\delta_{\mathbf{h}}(p)}{\delta_{\{0\}}(p)^k}.$$

By §5, we have $\delta_{\{0\}}(p) = 1$ for every odd split prime $p \nmid d$, so $\lambda_{\mathbf{h}}(p) = 1$ there for every $\mathbf{h}$ (§5 also shows $\delta_{\mathbf{h}}(p) = 1$ identically at split primes). Thus the only primes contributing a factor other than 1 are the odd inert primes ($\chi_d(p) = -1$, automatically $p \nmid d$ by definition of χ_d) together with the (finitely many, for fixed d) primes dividing d and $p = 2$. Convergence of the infinite product defining $\mathfrak{S}_{\mathbf{h}}$ is established in §7. For instance, for the tuple $\mathbf{h} = \{0, 1, 2\}$ and $d = -3$ considered above we have $\delta_{\mathbf{h}}(3) = 0$, so the factor at $p = 3$ vanishes and $\mathfrak{S}_{\mathbf{h}} = 0$. For such $\mathbf{h}$, no n has $n + h_1, \ldots, n + h_k$ all in E. We call such tuples inadmissible.

We are now in a position to state the hypothesis.

Hypothesis $U(k)$**.** *Fix $k \geq 1$. For every fixed $C > 0$:*

$$\max_{\mathbf{h}} \Big| \frac{\rho_k(\mathbf{h}; x)}{\rho_1(x)^k} - \mathfrak{S}_{\mathbf{h}} \Big| \ \longrightarrow \ 0 \qquad (x \to \infty),$$

the maximum being over all sets $\mathbf{h} \subset (0, C\sqrt{\log x}\,] \cap \mathbb{Z}$ of k distinct integers.

This is the uniform Hardy–Littlewood-type hypothesis on which Theorem 1.2 is conditional. It asserts that simultaneous representation of $n + h_1, \ldots, n + h_k$ behaves, to leading order, like k independent events each of probability $\rho_1(x)$, corrected by the local singular series, and that this holds not just for a fixed tuple but uniformly as the tuple ranges over an interval growing with x.

5 Local densities at odd primes: a unified recursive treatment

In this section we compute S_p and $\delta_{\{0\}}(p)$ exactly at every odd prime p, together with the bounds on $\delta_{\mathbf{h}}(p)$ needed in §7; apart from two standard facts quoted from [Cox13] and [Ser73], the treatment is self-contained. Fix an odd prime p. By Lemma 5.1 below, completing the square identifies S_p, up to a measure-preserving unit rescaling, with the image of the single quadratic map $Q(X, Y) = X^2 - dY^2$. The type of p — split, inert, or dividing d — corresponds to whether Q, reduced mod p, is isotropic, anisotropic, or degenerate, and this trichotomy is all that is needed to compute S_p exactly via one recursive mechanism.

5.1 Setup: completing the square

Lemma 5.1 (Leading coefficient; see [Cox13, Lemmas 2.3 and 2.25])**.** *Every primitive form of discriminant d is properly equivalent to one with leading coefficient prime to any given prime p.*

Fix such a representative $f = ax^2 + bxy + cy^2 \in R$, $p \nmid a$. Since p is odd, $2a$ is a unit, and completing the square gives:

$$4a \cdot f(x, y) = (2ax + by)^2 - dy^2.$$

Notice that the map $(x, y) \mapsto (X, Y) = (2ax + by, y)$ is a bijection $\mathbb{Z}_p^2 \to \mathbb{Z}_p^2$ (invert via $y = Y$, $x = (X - bY)/(2a)$), so that:

$$4a \cdot S_p = T_p := \{X^2 - dY^2 : \ X, Y \in \mathbb{Z}_p\}.$$

Since $4a$ is a unit, this rescaling preserves Haar measure *exactly at every finite level*: for any α, the map $n \mapsto 4an \bmod p^\alpha$ is a bijection of $\mathbb{Z}/p^\alpha\mathbb{Z}$, so $\#(A \bmod p^\alpha) = \#(4aA \bmod p^\alpha)$ for any set A, and taking $\alpha \to \infty$ we get $\mu_p(S_p) = \mu_p(T_p)$. It therefore suffices to compute T_p.

5.2 The nonsingular case: a single Hensel lemma

Lemma 5.2 (Nonsingular resolution)**.** *Let $(X_0, Y_0) \in \mathbb{Z}_p^2$ with $\nabla Q(X_0, Y_0) = (2X_0, -2dY_0) \not\equiv (0,0) \pmod{p}$, and suppose $Q(X_0, Y_0) \equiv m \pmod{p^\alpha}$ for some $\alpha \geq 1$. Then for every $m' \in m + p^\alpha \mathbb{Z}_p$ — every element of this coset, not merely m itself — the following holds. If $2X_0 \not\equiv 0 \pmod{p}$, there is a unique $X \equiv X_0 \pmod{p}$ with $Q(X, Y_0) = m'$ exactly, and it satisfies $X \equiv X_0 \pmod{p^\alpha}$. Similarly, if $2dY_0 \not\equiv 0 \pmod{p}$, there is a unique $Y \equiv Y_0 \pmod{p}$ with $Q(X_0, Y) = m'$ exactly, and it satisfies $Y \equiv Y_0 \pmod{p^\alpha}$.*

The idea of the proof is to run Newton's method for the quadratic polynomial $g(X) = Q(X, Y_0)$, starting from X_0, and to observe that, since g is quadratic, its Taylor expansion is exact, so that the precision doubles at every step.

Proof. Say $2X_0 \not\equiv 0 \pmod{p}$ (the case $2dY_0 \not\equiv 0$ is similar, with $-d$ in place of the leading coefficient 1). Fix $Y = Y_0$ and set $g(X) := Q(X, Y_0)$, so that $g'(X_0) = 2X_0 \in \mathbb{Z}_p^\times$. Given $m' \in m + p^\alpha \mathbb{Z}_p$, define a sequence by:

$$X_0 \text{ (given)}, \qquad X_{n+1} := X_n - \frac{g(X_n) - m'}{g'(X_n)}.$$

Since g is quadratic with leading coefficient 1, its Taylor expansion around any point terminates exactly at the quadratic term, with no remainder: $g(X_{n+1}) = g(X_n) + g'(X_n)(X_{n+1} - X_n) + (X_{n+1} - X_n)^2$. By definition of X_{n+1}, the first two terms on the right cancel $-(g(X_n) - m')$, leaving the exact identity:

$$g(X_{n+1}) - m' = (X_{n+1} - X_n)^2.$$

Claim 1: We have $X_n \equiv X_0 \pmod{p^\alpha}$ (in particular mod p) and $\beta_n := v_p(g(X_n) - m') \geq \alpha \cdot 2^n$, for every $n \geq 0$.

Proof of Claim 1: We argue by induction on n. For the base case, $X_0 \equiv X_0$ trivially, and $\beta_0 = v_p(Q(X_0, Y_0) - m') \geq \alpha$ is exactly the hypothesis (m' lying in the same coset as m). For the inductive step, assume the claim at n. Then $g'(X_n) = 2X_n$ is a unit (as p is odd and $X_n \equiv X_0 \pmod{p}$), so $v_p(X_{n+1} - X_n) = \beta_n - v_p(g'(X_n)) = \beta_n \geq \alpha \cdot 2^n \geq \alpha$. This gives $X_{n+1} \equiv X_n \pmod{p^\alpha}$, hence $X_{n+1} \equiv X_0 \pmod{p^\alpha}$ by the induction hypothesis on X_n. Moreover, the exact identity above gives $\beta_{n+1} = 2v_p(X_{n+1} - X_n) = 2\beta_n \geq \alpha \cdot 2^{n+1}$, which closes the induction. This proves Claim 1.

We now put Claim 1 to work. It gives $\beta_n \to \infty$, that is, $v_p(X_{n+1} - X_n) \to \infty$, so the sequence is Cauchy and converges to some $X \in \mathbb{Z}_p$ with $X \equiv X_0 \pmod{p^\alpha}$ (inherited directly from Claim 1, which holds at every n). Continuity then gives:

$$g(X) - m' = \lim_n (g(X_n) - m') = 0.$$

This proves existence.

Claim 2: The solution X is unique among the elements $\equiv X_0 \pmod{p}$.

Proof of Claim 2: Suppose $X'' \equiv X_0 \pmod{p}$ also satisfies $g(X'') = m'$. Then:

$$0 = g(X) - g(X'') = X^2 - X''^2 = (X - X'')(X + X'').$$

Since $X + X'' \equiv 2X_0 \pmod{p}$ is a unit (the nonsingularity hypothesis), hence nonzero, it must be the case that $X = X''$. This proves Claim 2 and completes the proof of the lemma. □

Remark 5.3. This is the special case of Newton's method where the Taylor expansion is exact rather than merely approximate — a genuine feature of Q being quadratic, not a general phenomenon — so the doubling of precision at each step ($\beta_{n+1} = 2\beta_n$ exactly) is established directly from the identity above, with no remainder term to bound.

5.3 The trichotomy mod p

The form $Q(X, Y) = X^2 - dY^2 \pmod{p}$ is a binary quadratic form over $\mathbb{F}_p$, and its type is read off directly from $\chi_d(p)$:

- $\chi_d(p) = 1$ (split): d is a nonzero square mod p, hence (Hensel; see [Ser73, Ch. II, §2.2]) a square in $\mathbb{Z}_p^\times$: $d = s^2$. Then $Q(X,Y) = (X - sY)(X + sY) = UV$ under the bijection $(X,Y) \mapsto (U,V) = (X - sY, X + sY)$ (invertible over $\mathbb{Z}_p$ since $2, s$ are units).
- $\chi_d(p) = -1$ (inert): d is a non-square unit, and Q is anisotropic: $Q(X,Y) \equiv 0 \pmod p$ forces $X \equiv Y \equiv 0 \pmod p$ (else $(X/Y)^2 \equiv d$, contradicting that d is a non-square).
- $\chi_d(p) = 0$ ($p \mid d$): writing $d = p^e\, d_0$ ($e = v_p(d) \geq 1$, $d_0 \in \mathbb{Z}_p^\times$), we have $Q(X,Y) \equiv X^2 \pmod p$ — degenerate, rank 1: the gradient's second component $-2dY$ vanishes mod p identically, for *every* Y, since $p \mid d$.

5.4 Split primes: resolved directly, no recursion

Proposition 5.4 (Split primes)**.** *If $\chi_d(p) = 1$, then $T_p = \mathbb{Z}_p$, so $\delta_{\{0\}}(p) = 1$.*

Proof. We have $T_p = \{UV : U, V \in \mathbb{Z}_p\}$. For any target m, take $U = 1$ and $V = m$. □

Since $S_p = \mathbb{Z}_p$ outright (Proposition 5.4), translating by any h_i leaves it unchanged: $S_p - h_i = \mathbb{Z}_p$. It follows that $\delta_{\mathbf{h}}(p) = 1$ identically at split primes, for every $\mathbf{h}$, not merely for $\mathbf{h} = \{0\}$. There is nothing left to compute, and split primes never enter the singular series (§4) at all.

5.5 Inert primes: a self-referential recursion

Proposition 5.5 (Inert primes)**.** *If $\chi_d(p) = -1$, then $T_p = \{0\} \cup \{m : v_p(m)$ even$\}$, and $\delta_{\{0\}}(p) = p/(p+1)$.*

Proof. Since Q is anisotropic mod p, $Q(X,Y) \equiv 0 \pmod p$ forces $X \equiv Y \equiv 0 \pmod p$. Moreover, every representative of a target $m \not\equiv 0 \pmod p$ is automatically nonsingular: if $X \equiv Y \equiv 0 \pmod p$ then $Q(X,Y) \equiv 0 \pmod p$, so a representative of a unit target has $(X,Y) \not\equiv (0,0) \pmod p$, and hence $\nabla Q = (2X, -2dY) \not\equiv 0 \pmod p$ since $p \nmid 2d$. A nondegenerate binary quadratic form over $\mathbb{F}_p$ represents every element of $\mathbb{F}_p$ (a classical counting fact), so every unit target has a nonsingular representative mod p, and by Lemma 5.2 the entire unit coset is then achieved. Thus $T_p \supseteq \mathbb{Z}_p^\times$, with $v_p(m) = 0$ fully represented.

Now let $v_p(m) \geq 1$. Only singular representatives are possible (shown above), that is, $X \equiv Y \equiv 0 \pmod p$. Writing $X = pX_1$ and $Y = pY_1$ we get:

$$m = p^2\big(X_1^2 - dY_1^2\big) = p^2\, Q(X_1, Y_1).$$

This is the identical problem, with X_1, Y_1 ranging freely over $\mathbb{Z}_p$, and it gives the exact recursive identity:

$$T_p = \mathbb{Z}_p^\times \;\cup\; p^2 T_p.$$

Unrolling, we find $T_p = \mathbb{Z}_p^\times \cup p^2\mathbb{Z}_p^\times \cup p^4\mathbb{Z}_p^\times \cup \cdots \cup \{0\} = \{0\} \cup \{v_p \text{ even}\}$, which proves the set identity. Taking measures, $\mu := \mu_p(T_p)$ satisfies $\mu = (1 - 1/p) + p^{-2}\mu$ (using $\mu_p(p^2 T_p) = p^{-2}\mu_p(T_p)$), so that:

$$\mu = \frac{1 - 1/p}{1 - 1/p^2} = \frac{1}{1 + 1/p} = \frac{p}{p+1}.$$

This completes the proof. □

5.6 Primes dividing d: the degenerate case, recursively

Write $d = p^e\, d_0$, $e = v_p(d) \geq 1$, $d_0 \in \mathbb{Z}_p^\times$. The idea is again to separate the nonsingular representatives, which are handled by Lemma 5.2, from the singular ones, which give back the same type of problem with the exponent e lowered by 2. This produces a recursion in e.

We start with the nonsingular representatives. Since $p \mid d$, the gradient's second component $-2dY$ vanishes mod p for *every* Y, so nonsingularity reduces to $X \not\equiv 0 \pmod p$, with Y entirely unconstrained. For such a representative, $v_p(dY^2) \geq e \geq 1$, so $Q(X,Y) \equiv X^2 \pmod p$ regardless of Y. Hence the achievable targets are exactly the units that are quadratic residues mod p: half the units, at $v_p = 0$, independent of e.

Consider next the case $e = 1$. Since $v_p(X^2)$ is always even and $v_p(dY^2) = 1+2v_p(Y)$ is always odd, these valuations can never coincide. No cancellation is ever possible, so $v_p(Q) = \min(2v_p(X),\, 1+2v_p(Y))$ exactly, for every $(X, Y) \neq (0, 0)$. Every valuation $j \geq 0$ is therefore achieved: we force $v_p(X) = j/2$ (with Y deep enough not to interfere) if j is even, or $v_p(Y) = (j-1)/2$ if j is odd, in each case achieving exactly half the residues at that level (a quadratic-residue-type condition on the surviving unit; the precise conditions are worked out in the proof of Lemma 5.12). Summing over j we obtain:

$$\mu_p(T_p) = \sum_{j\geq 0} \tfrac{1}{2}(1-\tfrac{1}{p})p^{-j} = \tfrac{1}{2}.$$

Now suppose $e \geq 2$. For a singular representative ($X \equiv 0 \pmod p$), write $X = pX_1$. Then:

$$Q(X, Y) = p^2X_1^2 - p^e\, d_0\, Y^2 = p^2\big(X_1^2 - p^{e-2}\, d_0\, Y^2\big) \qquad (e \geq 2).$$

This is exactly the same type of problem, with exponent $e-2$, the same d_0, and X_1, Y free. Together with the uniform nonsingular piece, it gives the exact recursive identity:

$$T_p^{(e)} = \{v_p = 0,\ \text{QR mod } p\} \ \cup\ p^2\, T_p^{(e-2)} \qquad (e \geq 2).$$

The recursion bottoms out, after $\lfloor e/2 \rfloor$ steps, at exactly one of three already-solved base cases: $T_p^{(1)}$ (the direct argument just given, if e is odd) or $T_p^{(0)}$ — the form $X^2 - d_0Y^2$ with $p \nmid d_0$, that is, Proposition 5.4 if $\left(\frac{d_0}{p}\right) = 1$ or Proposition 5.5 if $\left(\frac{d_0}{p}\right) = -1$ (if e is even).

Writing $\mu(e) := \mu_p(T_p^{(e)})$, the recursion reads:

$$\mu(e) = \tfrac{1}{2}(1-\tfrac{1}{p}) + p^{-2}\mu(e-2).$$

It has fixed point $L = \dfrac{p}{2(p+1)}$, obtained by solving $L = \frac{1}{2}(1-\frac{1}{p})+p^{-2}L$. Since $\mu(e)-L = p^{-2}(\mu(e-2)-L)$, iterating gives:

$$\mu(e) - L = p^{-2\lfloor e/2 \rfloor}(\mu(e_0) - L),$$

where $e_0 \in \{0, 1\}$ is the parity-matched base exponent. Computing $\mu(e_0) - L$ in each of the three cases yields the following three propositions.

Proposition 5.6 (Odd e)**.** *If $e \geq 1$ is odd, then:*

$$\delta_{\{0\}}(p) = \frac{p}{2(p+1)} + \frac{1}{2(p+1)}\, p^{-(e-1)}.$$

Proposition 5.7 (Even e, split base: $\left(\frac{d_0}{p}\right) = 1$)**.** *If $e \geq 2$ is even and d_0 is a square mod p, then:*

$$\delta_{\{0\}}(p) = \frac{p}{2(p+1)} + \frac{p+2}{2(p+1)}\, p^{-e}.$$

Proposition 5.8 (Even e, inert base: $\left(\frac{d_0}{p}\right) = -1$)**.** *If $e \geq 2$ is even and d_0 is not a square mod p, then:*

$$\delta_{\{0\}}(p) = \frac{p}{2(p+1)}\big(1 + p^{-e}\big).$$

Proof (of Propositions 5.6, 5.7, and 5.8). For e odd, we have $\mu(1) - L = \frac{1}{2} - \frac{p}{2(p+1)} = \frac{1}{2(p+1)}$, which gives Proposition 5.6 directly from $\mu(e) = L + p^{-(e-1)}(\mu(1) - L)$. For e even, we have $\mu(0) - L = 1 - \frac{p}{2(p+1)} = \frac{p+2}{2(p+1)}$ in the split case (Proposition 5.4), and $\mu(0) - L = \frac{p}{p+1} - \frac{p}{2(p+1)} = \frac{p}{2(p+1)}$ in the inert case (Proposition 5.5). Using $\mu(e) = L + p^{-e}(\mu(0) - L)$, this gives Propositions 5.7 and 5.8. This completes the proof. □

Remark 5.9. All three formulas converge to the same limit $p/(2(p+1))$ as $e \to \infty$ — exactly the density of squares in $\mathbb{Z}_p$. This is not a coincidence: the recursion's fixed point is forced by the nonsingular piece alone (which never depends on d_0 or the split/inert distinction), and as $e \to \infty$ the recursive piece $p^{-2\lfloor e/2 \rfloor}(\mu(e_0) - L)$ — the only place the base type enters — decays to 0, leaving only the type-independent fixed point.

5.7 Regularity, uniformly

The regularity lemmas below control how fast the truncated densities $\delta_{\mathbf{h}}^{(\alpha)}(p)$ approach $\delta_{\mathbf{h}}(p)$. They rest on a simple structural property of the sets S_p, namely that membership of a nonzero m is decided by m modulo a power of p only slightly larger than $p^{v_p(m)}$. This motivates the following definition.

Definition 5.10 (c-stable set)**.** Let p be a prime and $c \geq 1$. A set $A \subseteq \mathbb{Z}_p$ is *c-stable* if for every $m \in \mathbb{Z}_p \setminus \{0\}$ and every $m' \equiv m \pmod{p^{v_p(m)+c}}$ we have $m \in A \iff m' \in A$. In other words, for $m \neq 0$, membership of m in A is determined by m modulo $p^{v_p(m)+c}$.

The value 0 plays no role in this definition. Observe that if $m \neq 0$ and $m' \equiv m \pmod{p^{v_p(m)+c}}$, then $v_p(m') = v_p(m)$, so that the relation between m and m' is symmetric.

Lemma 5.11 (Closure properties of c-stable sets)**.** *Let $c \geq 1$.*

(a) Any union of c-stable subsets of $\mathbb{Z}_p$ is c-stable, and a c-stable set is c'-stable for every $c' \geq c$.

(b) If A is c-stable, then so is $p^2 A$.

(c) If A is c-stable and $\lambda \in \mathbb{Z}_p \setminus \{0\}$, then $\{m \in \mathbb{Z}_p : \lambda m \in A\}$ is c-stable.

(d) If $r \in \mathbb{Z}_p$ and $s \geq 1$ satisfy $s \leq v_p(r) + c$ (where $v_p(0) = \infty$), then the coset $r + p^s\mathbb{Z}_p$ is c-stable.

Proof. For (a), suppose $m \in \bigcup_i A_i$ and $m' \equiv m \pmod{p^{v_p(m)+c}}$. Then $m \in A_i$ for some i, so $m' \in A_i$ by c-stability of A_i; the converse follows from the symmetry of the relation. For the second assertion, observe that congruence modulo $p^{v_p(m)+c'}$ implies congruence modulo $p^{v_p(m)+c}$.

For (b), let $m \in p^2 A$ and $m' \equiv m \pmod{p^{v_p(m)+c}}$. Then $v_p(m') = v_p(m) \geq 2$, and $m'/p^2 \equiv m/p^2 \pmod{p^{v_p(m/p^2)+c}}$, so $m/p^2 \in A$ if and only if $m'/p^2 \in A$. This proves (b), since the case $v_p(m) < 2$ is trivial: neither m nor m' lies in $p^2 A$.

For (c), if $m' \equiv m \pmod{p^{v_p(m)+c}}$ then $\lambda m' \equiv \lambda m \pmod{p^{v_p(\lambda m)+c}}$, so $\lambda m \in A$ if and only if $\lambda m' \in A$.

For (d), write $U := r + p^s\mathbb{Z}_p$ and let $m' \equiv m \pmod{p^{v_p(m)+c}}$ with $m \neq 0$. If $v_p(r) < s$, then every element of U has valuation $v_p(r)$, so $s \leq v_p(m) + c$ whenever $m \in U$, and then $m' \equiv m \pmod{p^s}$ gives $m' \in U$. If $v_p(r) \geq s$, then $U = p^s\mathbb{Z}_p$ and the condition $m \in U$ means $v_p(m) \geq s$, which depends only on $v_p(m) = v_p(m')$. In either case the converse follows from symmetry. □

Lemma 5.12 (Stability of S_p at odd primes)**.** *For every odd prime p, the set S_p is 1-stable.*

Proof. By §5.1 we have $S_p = \{m : 4a\,m \in T_p\}$ with a a unit, so by Lemma 5.11(c) it is enough to show that T_p is 1-stable. If $\chi_d(p) = 1$ then $T_p = \mathbb{Z}_p$ (Proposition 5.4), and if $\chi_d(p) = -1$ then $T_p = \{0\} \cup \{v_p \text{ even}\}$ (Proposition 5.5). In both cases membership of $m \neq 0$ depends only on $v_p(m)$, so T_p is 1-stable.

Suppose now that $p \mid d$, write $d = p^e d_0$ as in §5.6, and write $T_p^{(e)}$ for T_p. We show by induction on e that $T_p^{(e)}$ is 1-stable. The base cases are $e = 0$, which is the case just treated (with d_0 in place of d), and $e = 1$. For $e \geq 2$, the recursion of §5.6 gives $T_p^{(e)} = \{v_p = 0,\ \text{QR mod } p\} \cup p^2 T_p^{(e-2)}$. The first set is a union of cosets $r + p\mathbb{Z}_p$ with r a unit, hence 1-stable by Lemma 5.11(d), and the second is 1-stable by Lemma 5.11(b) and the induction hypothesis. By Lemma 5.11(a) the union is 1-stable.

It remains to treat $e = 1$. As in §5.6, $v_p(Q(X,Y)) = \min(2v_p(X), 1 + 2v_p(Y))$ for $(X,Y) \neq (0,0)$. Let $m = p^j u$ with $u \in \mathbb{Z}_p^\times$. If $j = 2s$ is even, a representation $Q(X,Y) = m$ forces $v_p(X) = s$ and $v_p(Y) \geq s$, so that $u = X_0^2 - pd_0Y_1^2 \equiv X_0^2 \pmod{p}$ with $X_0 := X/p^s$ and $Y_1 := Y/p^s$. Hence u is a square mod p. Conversely, if u is a square mod p then it is a square in $\mathbb{Z}_p$ and $(X,Y) = (p^s\sqrt{u}, 0)$ works. If $j = 2s+1$ is odd, a representation forces $v_p(Y) = s$ and $v_p(X) \geq s+1$, so that $u = pX_1^2 - d_0Y_0^2 \equiv -d_0Y_0^2 \pmod{p}$ with $X_1 := X/p^{s+1}$ and $Y_0 := Y/p^s$, and hence $-u/d_0$ is a square mod p. Conversely, if $-u/d_0$ is a square mod p, then it is a square in $\mathbb{Z}_p$ and $(X,Y) = (0, p^s\sqrt{-u/d_0})$ works. In both cases, membership of m is determined by j and u mod p, that is, by m modulo p^{j+1}. This shows that $T_p^{(1)}$ is 1-stable and completes the proof. □

The next lemma turns stability into a bound on the truncation error for a single index.

Lemma 5.13 (Single-index truncation)**.** *Let $S \subseteq \mathbb{Z}_p$ be a closed c-stable set, and put $\delta := \mu_p(S)$ and $\delta^{(\alpha)} := p^{-\alpha}\,\#(S \bmod p^\alpha)$. Then for every $\alpha \geq 1$:*

$$0 \;\leq\; \delta^{(\alpha)} - \delta \;\leq\; \frac{p^{c-1}}{p^\alpha}.$$

Proof. Fix $\beta \geq 1$ and let a be a residue class modulo p^β that meets S, so that it is counted in $\delta^{(\beta)}$. Suppose first that $a \not\equiv 0 \pmod{p^\beta}$ and $v_p(a) \leq \beta - c$, where $v_p(a) < \beta$ denotes the valuation of any representative. Every $m \in a + p^\beta\mathbb{Z}_p$ then satisfies $v_p(m) = v_p(a)$, and $a + p^\beta\mathbb{Z}_p \subseteq m + p^{v_p(m)+c}\mathbb{Z}_p$. By c-stability, the whole class $a + p^\beta\mathbb{Z}_p$ is contained in S, since it meets S. Consequently all p of its refinements modulo $p^{\beta+1}$ are counted in $\delta^{(\beta+1)}$.

The remaining classes modulo p^β are those with $a \equiv 0$ or $v_p(a) \geq \beta - c + 1$, and there are at most p^{c-1} of them. Each of these classes meets S, so at least one of its p refinements modulo $p^{\beta+1}$ is counted in $\delta^{(\beta+1)}$, and at most $p-1$ refinements are lost, each of measure $p^{-(\beta+1)}$. Therefore:

$$0 \;\leq\; \delta^{(\beta)} - \delta^{(\beta+1)} \;\leq\; \frac{(p-1)\,p^{c-1}}{p^{\beta+1}}.$$

Since S is closed, $S = \bigcap_\beta \big(S + p^\beta\mathbb{Z}_p\big)$, and $\delta^{(\beta)} = \mu_p(S + p^\beta\mathbb{Z}_p)$ decreases to δ. Telescoping over $\beta \geq \alpha$ gives:

$$\delta^{(\alpha)} - \delta \;\leq\; p^{c-1} \sum_{\beta \geq \alpha} \frac{p-1}{p^{\beta+1}} = \frac{p^{c-1}}{p^\alpha},$$

and the lower bound holds because $\delta^{(\beta)}$ is decreasing in β. This completes the proof. □

With these two lemmas in hand, the regularity of the multi-index densities follows from a union bound.

Lemma 5.14 (Regularity at odd primes)**.** *For every finite $\mathbf{h} \subset \mathbb{Z}$ with $\#\mathbf{h} = k \geq 1$, every odd prime p, and every $\alpha \geq 1$:*

$$0 \;\leq\; \delta_{\mathbf{h}}^{(\alpha)}(p) - \delta_{\mathbf{h}}(p) \;\leq\; \frac{k}{p^\alpha}.$$

The core idea of the proof is to compare the truncated density with the true density through the union bound over the k shifts, which reduces the problem to the single-index discrepancy controlled by Lemma 5.13.

Proof. Throughout, $\delta_{\mathbf{h}}^{(\alpha)}(p) \geq \delta_{\mathbf{h}}(p)$: truncation only ever gains residues, never loses them, as the construction below makes precise. This gives the lower bound $0 \leq \delta_{\mathbf{h}}^{(\alpha)}(p) - \delta_{\mathbf{h}}(p)$ directly, so it is enough to prove the upper bound.

For $C \subseteq \mathbb{Z}_p$, write $\widehat{C}_\alpha$ for its coset-completion at level α, the union of residue classes mod p^α meeting C. Write $A := \bigcap_i B_i$ for $B_i := S_p - h_i$, so that $\delta_{\mathbf{h}}(p) = \mu_p(A)$, and write $\widetilde{A}_\alpha := \bigcap_i \widehat{B_i}_\alpha$ for the intersection of the *individual* completions.

Claim 1: We have $\delta_{\mathbf{h}}^{(\alpha)}(p) = \mu_p(\widetilde{A}_\alpha)$.

Proof of Claim 1: By Definition 4.4, the element a contributes to $\delta_{\mathbf{h}}^{(\alpha)}(p)$ exactly when $a + h_i$ lies in S_p mod p^α for *each* i separately, that is, when $a \in \widehat{B_i}_\alpha$ for every i. This proves Claim 1.

Observe that $\widetilde{A}_\alpha$ need not equal $\widehat{A}_\alpha$, the completion of the intersection, which can be genuinely smaller. For instance, at $\alpha = 1$ with several h_i coinciding mod p, a residue can pass every individual check while admitting no single consistent witness, so the distinction is not merely formal.

Claim 2: We have $\widetilde{A}_\alpha \setminus A \;\subseteq\; \bigcup_{i=1}^k \big(\widehat{B_i}_\alpha \setminus B_i\big)$.

Proof of Claim 2: Let $a \in \widetilde{A}_\alpha \setminus A$. Since $a \notin A = \bigcap_i B_i$, it must be the case that $a \notin B_{i_0}$ for some specific index i_0. On the other hand, $a \in \widetilde{A}_\alpha$ means directly that $a \in \widehat{B_{i_0}}_\alpha$, by definition of $\widetilde{A}_\alpha$ as the intersection over every index. Hence $a \in \widehat{B_{i_0}}_\alpha \setminus B_{i_0}$, which proves Claim 2.

We now put our claims to work. Since $A \subseteq \widetilde{A}_\alpha$ always (each $B_i \subseteq \widehat{B_i}_\alpha$), it suffices to bound $\mu_p(\widetilde{A}_\alpha \setminus A)$. By Claim 2 and subadditivity of μ_p (the union bound, applied here to a union of complements, not to the B_i's themselves), we get:

$$\mu_p(\widetilde{A}_\alpha \setminus A) \leq \sum_{i=1}^k \mu_p(\widehat{B_i}_\alpha \setminus B_i).$$

Each $B_i = S_p - h_i$ is a translate of S_p, so translation invariance of μ_p identifies $\mu_p(\widehat{B_{i\alpha}} \setminus B_i)$ with the single-index discrepancy $\delta^{(\alpha)}_{\{0\}}(p) - \delta_{\{0\}}(p)$. The set S_p is closed (it is compact, as noted before Definition 4.3), and it is 1-stable by Lemma 5.12. Lemma 5.13 with $c = 1$ therefore bounds this discrepancy by $1/p^\alpha$, regardless of h_i. Summing the k terms we find:

$$\delta^{(\alpha)}_{\mathbf{h}}(p) - \delta_{\mathbf{h}}(p) = \mu_p(\widetilde{A}_\alpha \setminus A) \;\le\; \sum_{i=1}^{k} \frac{1}{p^\alpha} = \frac{k}{p^\alpha}.$$

This completes the proof. □

5.8 Multi-point densities: near 1 away from $\det \mathbf{h}$

For the singular series to converge (§7), we need $\lambda_{\mathbf{h}}(p) = \delta_{\mathbf{h}}(p)/\delta_{\{0\}}(p)^k$ to be close to 1 for all but finitely many primes; the next two lemmas show this at the odd inert primes.

Lemma 5.15 (Near 1 at generic inert primes)**.** *Write* $\det(\mathbf{h}) := \prod_{i<j}(h_j - h_i)$*. If p is odd with $\chi_d(p) = -1$ and $p \nmid \det(\mathbf{h})$ — that is, $h_1, \dots, h_k$ are pairwise distinct mod p — then:*

$$\delta_{\mathbf{h}}(p) = 1 - \frac{k}{p+1}, \qquad \textit{so} \qquad \lambda_{\mathbf{h}}(p) = 1 + O_k(p^{-2}).$$

Proof. Since $h_1, \dots, h_k$ are pairwise distinct mod p, for any a at most one index i_0 can have $a + h_{i_0} \equiv 0 \pmod p$. The k events $\{a \equiv -h_i \pmod p\}$ are therefore pairwise disjoint, each of measure $1/p$. We split into two cases according to whether a lies in one of these events.

Case 1: The element a lies off these events, which has measure $1 - k/p$. Then every $a + h_i$ is a unit, hence automatically in S_p (Proposition 5.5: $v_p = 0$ is even). This gives a full contribution of $(1 - k/p) \times 1$.

Case 2: The element a lies on the event $a \equiv -h_{i_0} \pmod p$, which has measure $1/p$, for each of the k choices of i_0. Then every $a + h_j$ with $j \neq i_0$ remains a unit (as $h_j \not\equiv h_{i_0} \pmod p$), so those are automatically fine, and only $a + h_{i_0}$, now known to be $\equiv 0 \pmod p$, needs checking. Write $a + h_{i_0} = pc$: we need $v_p(a + h_{i_0}) = 1 + v_p(c)$ to be even, that is, $v_p(c)$ odd, and by Proposition 5.5 odd valuations have density exactly $1 - p/(p+1) = 1/(p+1)$. So this event contributes $\frac{1}{p} \cdot \frac{1}{p+1}$, for each of the k choices of i_0.

Summing the two cases we obtain:

$$\delta_{\mathbf{h}}(p) = \left(1 - \tfrac{k}{p}\right) + k \cdot \tfrac{1}{p} \cdot \tfrac{1}{p+1} = 1 - \tfrac{k}{p}\left(1 - \tfrac{1}{p+1}\right) = 1 - \frac{k}{p} \cdot \frac{p}{p+1} = 1 - \frac{k}{p+1}.$$

This completes the proof. □

Lemma 5.16 (Universal near 1 bound, no genericity assumed)**.** *If p is odd with $\chi_d(p) = -1$, then for every $\mathbf{h}$ (not just those with $p \nmid \det \mathbf{h}$):*

$$\lambda_{\mathbf{h}}(p) \;\le\; (1 + 1/p)^{k-1}.$$

Proof. We have $\delta_{\mathbf{h}}(p) \le \delta_{\{0\}}(p)$ trivially, since we are intersecting more conditions with a fixed one (using translation invariance of μ_p). Hence $\lambda_{\mathbf{h}}(p) \le \delta_{\{0\}}(p)^{1-k} = (1 + 1/p)^{k-1}$, by Proposition 5.5 and $k \ge 1$. □

Together, Lemmas 5.15 and 5.16 control every odd inert factor of the singular series: $\lambda_{\mathbf{h}}(p) = 1 + O_k(p^{-2})$ at all but the at most $\omega(\det \mathbf{h})$ inert primes dividing $\det(\mathbf{h})$, where Lemma 5.16 applies. Since $\lambda_{\mathbf{h}}(p) = 1$ at odd split primes, only the finitely many primes dividing $2d$ remain. Absolute convergence of the full product $\mathfrak{S}_{\mathbf{h}} = \prod_p \lambda_{\mathbf{h}}(p)$ also needs $\lambda_{\mathbf{h}}(p)$ to be bounded above at these primes (it is nonnegative by definition), which comes from the lower bounds on $\delta_{\{0\}}(p)$ in §5.6 and §6; these are combined in Corollary 7.2.

6 Local densities at $p = 2$: a parallel recursive treatment

In this section we compute S_2 and $\delta_{\{0\}}(2)$, running parallel to §5; apart from one standard fact quoted from [Ser73], the treatment is self-contained. At $p = 2$ the gradient $(2X, -2dY)$ of $Q(X, Y) = X^2 - dY^2$ vanishes mod 2 *identically*, so there is no analogue of "nonsingular mod p". The recursion survives, but is built instead around the parity of X and Y, reflecting that squares in $\mathbb{Z}_2^\times$ have index 4, not 2.

6.1 The key classical input

Lemma 6.1 (Squares of 2-adic units; see [Ser73, Ch. II, §3.3, Theorem 4])**.** $(\mathbb{Z}_2^\times)^2 = 1 + 8\mathbb{Z}_2$*, of index* 4 *in* $\mathbb{Z}_2^\times$*: every unit is* $\equiv 1, 3, 5,$ *or* 7 (mod 8)*, and is a square if and only if it is* $\equiv 1$ (mod 8)*.*

The proof of this classical result is beyond the scope of the present article; a detailed treatment can be found in [Ser73, Ch. II, §3.3]. We will use it repeatedly below.

6.2 Setup: two completing-the-square regimes

By Lemma 5.1, we may fix $f = ax^2 + bxy + cy^2 \in R$ with a odd.

Suppose first that $2 \nmid d$. Then necessarily $d \equiv 1 \pmod 4$, and b is odd since $d = b^2 - 4ac$. The substitution $X = 2ax + by$, $Y = y$ satisfies $X \equiv Y \pmod 2$ always (as b is odd), and it is only injective on $\mathbb{Z}_2^2$, not surjective: its image is exactly $\{(X, Y) : X \equiv Y \pmod 2\}$ (given such (X, Y), recover $y = Y$, $x = (X - bY)/(2a) \in \mathbb{Z}_2$, since the numerator is even and $2a$ has valuation exactly 1). So we get:

$$4a \cdot S_2 = T_2 := \{X^2 - dY^2 : \ X \equiv Y \pmod 2\}.$$

Here $4a$ has valuation 2, so it is not a unit, and this is a genuine $\times 4$ measure scaling rather than a preserving one. Since a itself is already a unit, we find:

$$\mu_2(T_2) = \mu_2(4a \cdot S_2) = \tfrac{1}{4}\mu_2(a \cdot S_2) = \tfrac{1}{4}\mu_2(S_2),$$

that is, $\mu_2(S_2) = 4\,\mu_2(T_2)$.

Suppose now that $4 \mid d$ (forced whenever $2 \mid d$, since $d \equiv 0, 1 \bmod 4$). Write $d = 4m$. Then $b = 2b'$ is forced, and $a \cdot f(x, y) = (ax + b'y)^2 - my^2$ (expand, using $m = b'^2 - ac$) gives a genuine bijection $\mathbb{Z}_2^2 \to \mathbb{Z}_2^2$. There is no parity obstruction this time, since only the unit a, not $2a$, is used, and we obtain:

$$a \cdot S_2 = T_2 := \{X^2 - mY^2 : \ X, Y \in \mathbb{Z}_2 \text{ unconstrained}\}, \qquad \mu_2(S_2) = \mu_2(T_2).$$

This is a genuine unit rescaling here, since a itself is odd.

The values of $X^2 - m_0Y^2$ with $v_2(m_0) \in \{0, 1\}$ will be needed repeatedly below, so we record their structure once. Call $(X, Y) \in \mathbb{Z}_2^2$ primitive if X, Y are not both even.

Lemma 6.2 (Primitive values)**.** *Let* $m_0 \in \mathbb{Z}_2$ *with* $v_2(m_0) \in \{0, 1\}$*, let* N_0 *be the set of values* $X^2 - m_0Y^2$ *at primitive* (X, Y)*, and let* N_{oo}, N_{oe}, N_{eo} *be the subsets of values at primitive pairs with* (X, Y) *both odd,* X *odd and* Y *even, and* X *even and* Y *odd, respectively. Then:*

(a) $\{X^2 - m_0Y^2 : \ X, Y \in \mathbb{Z}_2\} = \{0\} \cup \bigcup_{j \geq 0} 4^j N_0$*, and* $N_0 = N_{oo} \cup N_{oe} \cup N_{eo}$*.*

(b) $N_{oo} = (1 - m_0) + 8\mathbb{Z}_2$*.*

(c) If $v_2(m_0) = 0$*, then* $N_{oe} = 1 + 4\mathbb{Z}_2$ *and* $N_{eo} = -m_0 + 4\mathbb{Z}_2$*.*

(d) If $v_2(m_0) = 1$*, then* $N_{oe} = 1 + 8\mathbb{Z}_2$ *and* $N_{eo} = \big(-m_0 + 16\mathbb{Z}_2\big) \cup \big(4 - m_0 + 16\mathbb{Z}_2\big)$*.*

Proof. For part (a), a pair $(X, Y) \neq (0, 0)$ can be written uniquely as $2^j(X_0, Y_0)$ with $j = \min(v_2(X), v_2(Y))$ and (X_0, Y_0) primitive, and then $X^2 - m_0Y^2 = 4^j(X_0^2 - m_0Y_0^2)$. Conversely, every $4^j n_0$ with $n_0 \in N_0$ arises this way. The value 0 arises at $(0, 0)$. The second assertion is the definition of the three subsets.

For parts (b)–(d) recall from Lemma 6.1 that the squares of odd elements of $\mathbb{Z}_2$ are exactly the elements of $1 + 8\mathbb{Z}_2$.

For part (b), if X, Y are both odd, then X^2, Y^2 range independently over $1 + 8\mathbb{Z}_2$, so $X^2 - m_0Y^2$ lies in:

$$(1 + 8\mathbb{Z}_2) - m_0(1 + 8\mathbb{Z}_2) = (1 - m_0) + 8\mathbb{Z}_2,$$

because $8m_0\mathbb{Z}_2 \subseteq 8\mathbb{Z}_2$. Every element of this coset occurs already with $Y = 1$ and X^2 varying.

For part (c), let m_0 be odd. If X is odd and $Y = 2w$ is even, then $X^2 - m_0Y^2 = X^2 - 4m_0w^2$, and $4m_0w^2 \equiv 0 \pmod 8$ for w even while $4m_0w^2 \equiv 4 \pmod 8$ for w odd. Hence this value lies in $(1+8\mathbb{Z}_2)\cup(5+8\mathbb{Z}_2) = 1 + 4\mathbb{Z}_2$, and both cosets occur (take $w = 0$, respectively $w = 1$ with X^2 varying). If $X = 2w$ is even and Y is odd, then $X^2 - m_0Y^2 = 4w^2 - m_0Y^2$, where m_0Y^2 ranges over $m_0 + 8\mathbb{Z}_2$, and $4w^2 \equiv 0 \pmod 8$ for w even while $4w^2 \equiv 4 \pmod 8$ for w odd. Hence the value lies in $(-m_0+8\mathbb{Z}_2)\cup(4-m_0+8\mathbb{Z}_2) = -m_0+4\mathbb{Z}_2$, and both cosets occur (take $w = 0$, respectively $w = 1$, with Y^2 varying).

For part (d), let $v_2(m_0) = 1$. If X is odd and $Y = 2w$ is even, then $X^2 - m_0Y^2 = X^2 - 4m_0w^2$ with $4m_0w^2 \equiv 0 \pmod 8$, so this value lies in $1 + 8\mathbb{Z}_2$, and all of $1 + 8\mathbb{Z}_2$ occurs with $w = 0$. If $X = 2w$ is even and Y is odd, then $X^2 - m_0Y^2 = 4w^2 - m_0Y^2$. Here Y^2 ranges over $1+8\mathbb{Z}_2$ by Lemma 6.1, so m_0Y^2 ranges over $m_0+8m_0\mathbb{Z}_2 = m_0+16\mathbb{Z}_2$, and $4w^2 \equiv 0 \pmod{16}$ for w even while $4w^2 \equiv 4 \pmod{16}$ for w odd. Hence the value lies in $(-m_0 + 16\mathbb{Z}_2) \cup (4 - m_0 + 16\mathbb{Z}_2)$, and both cosets occur (take $w = 0$, respectively $w = 1$, with Y^2 varying). This completes the proof. □

6.3 The case $2 \nmid d$: split and inert at 2

Suppose first that $d \equiv 1 \bmod 8$ (split). By Lemma 6.1, $d = s^2$ for a unit s, so Q factors as $X^2 - dY^2 = (X - sY)(X + sY)$. This substitution has determinant $2s$ (valuation exactly 1, not a unit), so it is injective but not surjective on $\mathbb{Z}_2^2$. Tracing parities directly, we see that $X \equiv Y \bmod 2$ corresponds exactly to U, V both even.

Proposition 6.3 (Split at 2)**.** *$T_2 = 4\mathbb{Z}_2$, so $\mu_2(S_2) = 1$.*

Proof. By the parity correspondence above, $T_2 = \{UV : U, V \text{ both even}\}$. Writing $U = 2U'$ and $V = 2V'$, with U', V' free since "both even" is the only constraint, we get $UV = 4U'V'$. As U', V' range unconstrained over $\mathbb{Z}_2$, the product $U'V'$ hits every target directly (for example, $U' = 1, V' = n$ gives n), so $T_2 = 4\mathbb{Z}_2$ exactly. This gives $\mu_2(T_2) = 1/4$, and by the scaling of §6.2 we conclude that $\mu_2(S_2) = 4 \cdot \frac{1}{4} = 1$. □

Suppose now that $d \equiv 5 \bmod 8$ (inert). Then d is not a square (Lemma 6.1), so the factorization trick is unavailable, and we argue by valuation directly.

Proposition 6.4 (Inert at 2)**.** *If $d \equiv 5 \pmod 8$, then $T_2 = \{0\} \cup \{n : v_2(n) \text{ even}, v_2(n) \geq 2\}$, so $\mu_2(S_2) = 2/3$.*

Proof. Let $(X, Y) \neq (0, 0)$ with $X \equiv Y \pmod 2$, and let $j = \min(v_2(X), v_2(Y))$. We consider the cases $j = 0$ and $j \geq 1$ separately.

If $j = 0$, then X, Y are both odd, and by Lemma 6.2(b) the values $X^2 - dY^2$ fill exactly the coset $(1 - d) + 8\mathbb{Z}_2 = 4 + 8\mathbb{Z}_2$, which is the set of elements of valuation exactly 2.

If $j \geq 1$, then X, Y are both even and the parity condition is automatic: $(X, Y) = 2^j(X_0, Y_0)$ with (X_0, Y_0) an arbitrary primitive pair, and $X^2 - dY^2 = 4^j n_0$ with $n_0 \in N_0$, where N_0 is the set of Lemma 6.2 for $m_0 = d$. Since $d \equiv 5 \pmod 8$ is odd, parts (b) and (c) of that lemma give:

$$N_{oo} = 4 + 8\mathbb{Z}_2, \qquad N_{oe} = 1 + 4\mathbb{Z}_2, \qquad N_{eo} = -d + 4\mathbb{Z}_2 = 3 + 4\mathbb{Z}_2,$$

so that $N_0 = \mathbb{Z}_2^\times \cup (4 + 8\mathbb{Z}_2)$, the union of the set of elements of valuation 0 and the set of elements of valuation 2. Therefore $4^j N_0$ is the union of the elements of valuation $2j$ and those of valuation $2j + 2$.

Putting the two cases together we find:

$$T_2 = \{0\} \ \cup \ \{v_2 = 2\} \ \cup \ \bigcup_{j \geq 1} \big(\{v_2 = 2j\} \cup \{v_2 = 2j + 2\}\big) = \{0\} \cup \{n : \ v_2(n) \text{ even}, \ v_2(n) \geq 2\}.$$

The set $\{v_2 = 2j\}$ has measure $\frac{1}{2} \cdot 4^{-j}$, so $\mu_2(T_2) = \sum_{j \geq 1} \frac{1}{2} \cdot 4^{-j} = \frac{1}{6}$ and $\mu_2(S_2) = 4 \cdot \frac{1}{6} = \frac{2}{3}$. This completes the proof. □

Thus, for $2 \nmid d$, the prime 2 behaves like a split or an inert prime in disguise: for $d \equiv 1 \bmod 8$ we have $\delta_{\{0\}}(2) = 1$ and 2 contributes nothing to the singular series, while for $d \equiv 5 \bmod 8$ we have $\delta_{\{0\}}(2) = 2/3$, matching the shape $p/(p + 1)$ at "$p = 2$" although it is obtained by mod-8 bookkeeping.

6.4 The case $4 \mid d$: four base types, then a recursion in the conductor

Write $d = 4m$. By §6.2, $\mu_2(S_2) = \mu_2(T_2)$ for $T_2 = \{X^2 - mY^2 : X, Y \in \mathbb{Z}_2\}$, unconstrained. Write $m = 4^\kappa m_0$ with $\kappa \geq 0$ and $v_2(m_0) \in \{0, 1\}$, and set $T_2^{(\kappa)} := \{X^2 - 4^\kappa m_0 Y^2\}$, so that $T_2 = T_2^{(\kappa)}$.

We first treat the four base types ($\kappa = 0$), that is, $T_2^{(0)} = \{X^2 - m_0 Y^2 : X, Y \in \mathbb{Z}_2\}$ with $v_2(m_0) \in \{0, 1\}$. By Lemma 6.2(a), $T_2^{(0)} = \{0\} \cup \bigcup_{j \geq 0} 4^j N_0$, so it suffices to determine N_0 in each case. We write $\mu_2(S_2) = \mu_2(T_2^{(0)})$ throughout (§6.2).

Proposition 6.5 (Base type $m_0 \equiv 1 \bmod 8$)**.** *For $m_0 \equiv 1 \bmod 8$: $T_2^{(0)} = \mathbb{Z}_2 \setminus \{v_2 = 1\}$, $\mu_2(S_2) = 3/4$.*

Proof. Since $1 - m_0 \equiv 0 \pmod 8$, Lemma 6.2(b) gives $N_{oo} = 8\mathbb{Z}_2$, and part (c) gives $N_{oe} = 1 + 4\mathbb{Z}_2$ and $N_{eo} = -m_0 + 4\mathbb{Z}_2 = 3 + 4\mathbb{Z}_2$. Thus $N_0 = \mathbb{Z}_2^\times \cup 8\mathbb{Z}_2$: all units, together with all elements of valuation at least 3, and 0. Then $j = 0$ contributes every unit and every element of valuation ≥ 3, and $j = 1$ contributes $4\mathbb{Z}_2^\times = \{v_2 = 2\}$. Together these (and 0) exhaust $\mathbb{Z}_2 \setminus \{v_2 = 1\}$. Conversely, an element of $4^j N_0$ has valuation $2j$ or at least $2j + 3$, which is never 1. Hence $T_2^{(0)} = \mathbb{Z}_2 \setminus \{v_2 = 1\}$, and $\mu_2(\{v_2 = 1\}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ gives $\mu_2(S_2) = \frac{3}{4}$. □

Proposition 6.6 (Base type $m_0 \equiv 5 \bmod 8$)**.** *For $m_0 \equiv 5 \bmod 8$: $\mu_2(S_2) = 2/3$.*

Proof. Since $1 - m_0 \equiv 4 \pmod 8$, Lemma 6.2 gives $N_{oo} = 4 + 8\mathbb{Z}_2$, $N_{oe} = 1 + 4\mathbb{Z}_2$, and $N_{eo} = -m_0 + 4\mathbb{Z}_2 = 3 + 4\mathbb{Z}_2$, so $N_0 = \mathbb{Z}_2^\times \cup (4 + 8\mathbb{Z}_2)$, the elements of valuation 0 and those of valuation 2. As in the proof of Proposition 6.4, $4^j N_0$ consists of the elements of valuation $2j$ and $2j + 2$, so $T_2^{(0)} = \{0\} \cup \{v_2 \text{ even}\}$ and $\mu_2(T_2^{(0)}) = \sum_{j \geq 0} \frac{1}{2} \cdot 4^{-j} = \frac{2}{3}$. □

Proposition 6.7 (Base type $m_0 \equiv 3 \bmod 4$)**.** *For $m_0 \equiv 3, 7 \bmod 8$ (i.e. m_0 odd, $m_0 \equiv 3 \bmod 4$): $\mu_2(S_2) = 1/2$.*

Proof. Here $-m_0 \equiv 1 \pmod 4$, so Lemma 6.2(c) gives $N_{oe} = N_{eo} = 1 + 4\mathbb{Z}_2$, and part (b) gives $N_{oo} = (1 - m_0) + 8\mathbb{Z}_2$, a single coset mod 8 of elements of valuation exactly 1 (as $1 - m_0 \equiv 2 \pmod 4$). So N_0 is the disjoint union of $1 + 4\mathbb{Z}_2$, of measure $\frac{1}{4}$ and consisting of units, and $(1 - m_0) + 8\mathbb{Z}_2$, of measure $\frac{1}{8}$ and consisting of elements of valuation 1. Hence $4^j N_0$ consists of elements of valuations $2j$ and $2j + 1$ only, the sets $4^j N_0$ for different j are pairwise disjoint, and $\mu_2(4^j N_0) = 4^{-j} \cdot \frac{3}{8}$. Therefore $\mu_2(T_2^{(0)}) = \sum_{j \geq 0} 4^{-j} \cdot \frac{3}{8} = \frac{3}{8} \cdot \frac{4}{3} = \frac{1}{2}$. □

Proposition 6.8 (Base type $v_2(m_0) = 1$)**.** *For $v_2(m_0) = 1$: $\mu_2(S_2) = 1/2$.*

Proof. By Lemma 6.2(b) and (d), N_0 is the union of the two sets $(1 + 8\mathbb{Z}_2) \cup \big((1 - m_0) + 8\mathbb{Z}_2\big)$ and $(-m_0 + 16\mathbb{Z}_2) \cup (4 - m_0 + 16\mathbb{Z}_2)$. The first consists of units, and its two cosets are distinct because $1 - m_0 \equiv 3 \pmod 4$; it has measure $\frac{1}{8} + \frac{1}{8} = \frac{1}{4}$. The second consists of elements of valuation exactly 1, and its two cosets mod 16 are distinct (they differ by 4); it has measure $\frac{1}{16} + \frac{1}{16} = \frac{1}{8}$. As in the previous proof, the sets $4^j N_0$ are pairwise disjoint, with $\mu_2(4^j N_0) = 4^{-j} \cdot \frac{3}{8}$, so $\mu_2(T_2^{(0)}) = \frac{3}{8} \cdot \frac{4}{3} = \frac{1}{2}$. □

With the four base types in hand, we now derive the recursion for $\kappa \geq 1$. We split on the parity of X alone.

If X is odd (any Y), then $v_2(4^\kappa m_0 Y^2) = 2\kappa + v_2(m_0) + 2v_2(Y) \geq 2\kappa \geq 2$, while $v_2(X^2) = 0$. These can never coincide, so $v_2(Q) = 0$ unconditionally. For $\kappa \geq 2$, we have $4^\kappa m_0 Y^2 \equiv 0 \pmod 8$ regardless of Y (as $2\kappa \geq 4$), so $Q \equiv X^2 \equiv 1 \pmod 8$: the full coset $1 + 8\mathbb{Z}_2$, of density $1/8$, with no ambiguity (every element of $1 + 8\mathbb{Z}_2$ is a square by Lemma 6.1, so every such value is attained already with $Y = 0$). For $\kappa = 1$, an identical computation splits on the parity of Y, giving $1 + 8\mathbb{Z}_2$ (if Y is even) or $(1 - 4m_0) + 8\mathbb{Z}_2$ (if Y is odd). For m_0 odd, $4m_0 \equiv 4 \bmod 8$ always, so this is $\{1, 5\} \bmod 8$ (density $1/4$); for $v_2(m_0) = 1$, $4m_0 \equiv 0 \bmod 8$, so both parities give $1 + 8\mathbb{Z}_2$ (density $1/8$).

If X is even, then $X = 2X_1$, and we find:

$$Q = 4X_1^2 - 4 \cdot 4^{\kappa-1} m_0 Y^2 = 4\big(X_1^2 - 4^{\kappa-1} m_0 Y^2\big) = 4\, Q_{(\kappa-1)}(X_1, Y).$$

This is the identical problem, one level down in κ, with X_1, Y free.

Proposition 6.9 (The recursion at 2, solved)**.** *Writing B_κ for the (X-odd) piece above and $\nu(\kappa) := \mu_2(T_2^{(\kappa)})$, the recursion $\nu(\kappa) = \mu_2(B_\kappa) + \frac{1}{4}\nu(\kappa-1)$ holds for $\kappa \geq 1$, with $\mu_2(B_\kappa) = 1/8$ for $\kappa \geq 2$ and $\mu_2(B_1) = 1/4$ (odd m_0) or $1/8$ (even m_0). Solving: the fixed point of the $\kappa \geq 2$ recursion is $L = 1/6$, and:*

$$\nu(\kappa) = \frac{1}{6} + \beta \cdot 4^{-\kappa}, \qquad \beta = 4\Big(\nu(1) - \frac{1}{6}\Big), \qquad \kappa \geq 1,$$

with $\nu(1) = \mu_2(B_1) + \frac{1}{4}\nu(0)$ computed from Propositions 6.5–6.8, giving:

$$\beta = \begin{cases} 13/12 & m_0 \equiv 1 \bmod 8 \ (\textit{split base}) \\ 1 & m_0 \equiv 5 \bmod 8 \ (\textit{inert base}) \\ 5/6 & m_0 \equiv 3, 7 \bmod 8 \ (\textit{ramified-odd base}) \\ 1/3 & v_2(m_0) = 1 \ (\textit{even-}m_0 \textit{ base}) \end{cases}.$$

Proof. We first treat $\kappa \geq 2$. Substituting $\mu_2(B_\kappa) = 1/8 = \frac{1}{6}\cdot\frac{3}{4}$ and solving, we get $\nu(\kappa) - \frac{1}{6} = \frac{1}{4}\big(\nu(\kappa-1) - \frac{1}{6}\big)$, so that $\nu(\kappa) - \frac{1}{6} = 4^{-(\kappa-1)}(\nu(1) - \frac{1}{6})$, that is, $\nu(\kappa) = \frac{1}{6} + \beta \cdot 4^{-\kappa}$ with $\beta = 4(\nu(1) - \frac{1}{6})$, for every $\kappa \geq 1$ (the formula for $\kappa = 1$ recovers itself trivially). It remains to compute $\nu(1)$ in each of the four cases.

Case 1: Split base. We have $\nu(1) = \frac{1}{4} + \frac{1}{4}\cdot\frac{3}{4} = \frac{7}{16}$, hence $\beta = 4(\frac{7}{16} - \frac{1}{6}) = 4 \cdot \frac{13}{48} = \frac{13}{12}$.

Case 2: Inert base. We have $\nu(1) = \frac{1}{4} + \frac{1}{4}\cdot\frac{2}{3} = \frac{5}{12}$, hence $\beta = 4(\frac{5}{12} - \frac{1}{6}) = 4 \cdot \frac{1}{4} = 1$.

Case 3: Ramified-odd base. We have $\nu(1) = \frac{1}{4} + \frac{1}{4}\cdot\frac{1}{2} = \frac{3}{8}$, hence $\beta = 4(\frac{3}{8} - \frac{1}{6}) = 4 \cdot \frac{5}{24} = \frac{5}{6}$.

Case 4: Even-m_0 base. We have $\nu(1) = \frac{1}{8} + \frac{1}{4}\cdot\frac{1}{2} = \frac{1}{4}$, hence $\beta = 4(\frac{1}{4} - \frac{1}{6}) = 4 \cdot \frac{1}{12} = \frac{1}{3}$.

This completes the proof. □

Since $\mu_2(S_2) = \mu_2(T_2)$ in this regime (§6.2, genuine unit rescaling), Proposition 6.9 gives $\delta_{\{0\}}(2) = \nu(\kappa) = \frac{1}{6} + \beta \cdot 4^{-\kappa}$ directly, for every conductor depth $\kappa \geq 1$. This completes the exact computation of $\delta_{\{0\}}(2)$ in every case.

6.5 Regularity at $p = 2$

At $p = 2$ the membership of m in S_2 is no longer decided by m modulo $2^{v_2(m)+1}$, since squares of units are detected modulo 8 and not modulo 2. It is, however, decided by m modulo $2^{v_2(m)+3}$, in the sense of Definition 5.10.

Lemma 6.10 (Stability of S_2)**.** *The set S_2 is 3-stable.*

Proof. By §6.2 we have $S_2 = \{m : \ \lambda m \in T_2\}$ with $\lambda = 4a$ if $2 \nmid d$ and $\lambda = a$ if $4 \mid d$, so by Lemma 5.11(c) it is enough to show that T_2 is 3-stable.

Suppose first that $2 \nmid d$. By Propositions 6.3 and 6.4, T_2 is either $4\mathbb{Z}_2$ or $\{0\} \cup \{n : \ v_2(n) \text{ even}, \ v_2(n) \geq 2\}$. In both cases membership of $n \neq 0$ depends only on $v_2(n)$, so T_2 is 1-stable, and hence 3-stable by Lemma 5.11(a).

Suppose now that $4 \mid d$, and write $T_2 = T_2^{(\kappa)}$ as in §6.4. We argue by induction on κ. Let $\kappa = 0$. By Lemma 6.2(a), $T_2^{(0)} = \{0\} \cup \bigcup_{j\geq 0} 4^j N_0$ and $N_0 = N_{oo} \cup N_{oe} \cup N_{eo}$. By parts (b)–(d) of that lemma, each of N_{oo}, N_{oe}, and N_{eo} is a union of cosets $r + 2^s\mathbb{Z}_2$ of one of the following kinds:

- cosets of $8\mathbb{Z}_2$ or $4\mathbb{Z}_2$ with r a unit, so that $s \leq 3 = v_2(r) + 3$;
- the coset $(1 - m_0) + 8\mathbb{Z}_2$ when $v_2(m_0) = 0$, where $v_2(1 - m_0) \geq 1$, so that $s = 3 \leq v_2(1 - m_0) + 3$;
- the cosets $-m_0 + 16\mathbb{Z}_2$ and $4 - m_0 + 16\mathbb{Z}_2$ when $v_2(m_0) = 1$, both of whose representatives have valuation exactly 1, so that $s = 4 = 1 + 3$.

Each of these is 3-stable by Lemma 5.11(d), so N_0 is 3-stable by Lemma 5.11(a). Applying Lemma 5.11(b) j times, each $4^j N_0$ is 3-stable, and so is their union. This proves the case $\kappa = 0$.

Let $\kappa \geq 1$. By the analysis of §6.4, the values of Q at pairs with X odd are units and form a set B_κ, while those at pairs with X even form $4T_2^{(\kappa-1)}$. Hence $T_2^{(\kappa)} = B_\kappa \cup 4T_2^{(\kappa-1)}$, where B_κ is a union of cosets of $8\mathbb{Z}_2$ consisting of units ($1 + 8\mathbb{Z}_2$, or $\{1, 5\} \bmod 8$, or $(1 - 4m_0) + 8\mathbb{Z}_2$). Each such coset is 3-stable by Lemma 5.11(d), and $4T_2^{(\kappa-1)}$ is 3-stable by Lemma 5.11(b) and the induction hypothesis. By Lemma 5.11(a) their union is 3-stable. This completes the proof. □

Lemma 6.11 (Regularity at $p = 2$)**.** *For every finite $\mathbf{h} \subset \mathbb{Z}$ with $\#\mathbf{h} = k \geq 1$ and every $\alpha \geq 1$:*

$$0 \;\leq\; \delta_{\mathbf{h}}^{(\alpha)}(2) - \delta_{\mathbf{h}}(2) \;\leq\; \frac{4k}{2^\alpha}.$$

Proof. The proof is identical to that of Lemma 5.14, with Lemma 6.10 in place of Lemma 5.12. The set S_2 is closed and 3-stable, so Lemma 5.13 with $p = 2$ and $c = 3$ bounds the single-index discrepancy $\delta_{\{0\}}^{(\alpha)}(2) - \delta_{\{0\}}(2)$ by $2^{c-1}/2^\alpha = 4/2^\alpha$. The reduction to the single-index discrepancy by the union bound over the k shifts (Claims 1 and 2 in the proof of Lemma 5.14) does not use the parity of p, and gives the stated bound. □

7 The singular series and its average

In this section we show that the product $\mathfrak{S}_{\mathbf{h}} = \prod_p \lambda_{\mathbf{h}}(p)$ converges, and that $\mathfrak{S}_{\mathbf{h}}$ has the correct average over a window of shifts. For the latter we adapt to our setting the simplified argument of Ford [For16], which replaces Gallagher's original proof [Gal76] for the primes; the exact local densities and regularity lemmas of §5 and §6 replace the corresponding facts for the primes.

7.1 A Fubini/CRT averaging identity

Lemma 7.1 (Averaging identity)**.** *Let G be a finite abelian group, $T \subseteq G$ any subset, $\mu(T) = \#T/\#G$. For $k \geq 1$,*

$$\frac{1}{\#G^k} \sum_{\mathbf{h} \in G^k} \mu\Big(\bigcap_i (T - h_i) \Big) = \mu(T)^k.$$

Proof. We expand the left-hand side:

$$\frac{1}{\#G^k} \sum_{\mathbf{h}} \frac{1}{\#G} \#\{a \in G : \; a + h_i \in T \;\forall i\} = \frac{1}{\#G^{k+1}} \sum_{\mathbf{h}} \sum_a \prod_i \mathbf{1}_T(a + h_i).$$

Now swap the order of summation. For fixed a, we have:

$$\sum_{\mathbf{h} \in G^k} \prod_i \mathbf{1}_T(a + h_i) = \prod_i \Big(\sum_{h_i \in G} \mathbf{1}_T(a + h_i) \Big) = (\#T)^k,$$

since each factor sums to $\#T$ independently, as $h_i \mapsto a + h_i$ is a bijection of G. So the total is:

$$\frac{1}{\#G^{k+1}} \sum_a (\#T)^k = \frac{\#G \cdot (\#T)^k}{\#G^{k+1}} = \mu(T)^k,$$

which is exactly what we wanted to show. □

Apply this with $G = \mathbb{Z}/p^\alpha\mathbb{Z}$ and T the set defining $\delta_{\mathbf{h}}^{(\alpha)}(p)$ at each prime (Definition 4.4). Since Lemma 7.1 holds for *any* subset T, it applies verbatim, and gives:

$$\frac{1}{p^{k\alpha}} \sum_{\mathbf{h} \bmod p^\alpha} \lambda_{\mathbf{h}}^{(\alpha)}(p) = 1, \qquad \lambda_{\mathbf{h}}^{(\alpha)}(p) := \frac{\delta_{\mathbf{h}}^{(\alpha)}(p)}{\delta_{\{0\}}^{(\alpha)}(p)^k}.$$

where k is the length of the tuple $\mathbf{h}$, even when its entries repeat. Thus the truncated local factor averages to exactly 1, at every prime and every level, regardless of what S_p actually is.

7.2 Convergence of the singular series

Corollary 7.2 (Convergence of the singular series)**.** *For every finite* $\mathbf{h} \subset \mathbb{Z}$ *with* $\#\mathbf{h} = k$*, the product* $\mathfrak{S}_{\mathbf{h}} = \prod_p \lambda_{\mathbf{h}}(p)$ *converges absolutely, and:*

$$\mathfrak{S}_{\mathbf{h}} \ \ll_{k,d} \ \big(\log\log(|\det \mathbf{h}| + 3)\big)^{k-1}.$$

If $\lambda_{\mathbf{h}}(p) = 0$ *for some* p*, then we set* $\mathfrak{S}_{\mathbf{h}} = 0$*, and absolute convergence refers to the remaining factors.*

Proof. At an odd split prime we have $S_p = \mathbb{Z}_p$ (Proposition 5.4), so $\lambda_{\mathbf{h}}(p) = 1$ and these primes may be ignored. We split the rest of the product into four kinds of primes, which are clearly disjoint (by parity, and by the trichotomy of §5.3): $p = 2$, the odd primes dividing d, the odd inert primes dividing $\det(\mathbf{h})$ (at most $\omega(\det \mathbf{h})$ of them), and the remaining odd inert primes. The first two kinds together are exactly the primes dividing $2d$, so there are $r := \omega(2d)$ of them. We treat three cases.

Case 1: The first two kinds, that is, $p = 2$ and the odd primes dividing d. Here $\delta_{\{0\}}(p) \geq 1/6$ uniformly. Indeed, Propositions 5.6–5.8 give $\delta_{\{0\}}(p) > p/(2(p+1)) \geq 3/8$ for odd $p \mid d$, while for $p = 2$ Propositions 6.3, 6.4, and 6.5–6.8 give $\delta_{\{0\}}(2) \geq 1/2$ when $2 \nmid d$ or $4 \mid d$ with conductor depth $\kappa = 0$, and Proposition 6.9 gives $\delta_{\{0\}}(2) > 1/6$ for every conductor depth $\kappa \geq 1$. Now $\delta_{\mathbf{h}}(p) \leq \delta_{\{0\}}(p)$, by translation invariance of μ_p and since we are intersecting more conditions with a fixed one, so we get $\lambda_{\mathbf{h}}(p) \leq \delta_{\{0\}}(p)^{1-k} \leq 6^{k-1}$.

Case 2: The odd inert primes dividing $\det(\mathbf{h})$. Here Lemma 5.16 gives $\lambda_{\mathbf{h}}(p) \leq (1 + 1/p)^{k-1}$. The number of such primes is at most $\omega(n)$, where $n := |\det \mathbf{h}|$, and is not bounded in terms of k alone, so we must bound the product of these factors with some care. A product $\prod_{p \in P}(1 + 1/p)$ over a set P of $\omega(n)$ primes is largest when P consists of the first $\omega(n)$ primes. Since $\omega(n) \ll \log n / \log\log n$, the $\omega(n)$-th prime is $\ll \log n$, and Mertens' second theorem gives:

$$\prod_{\substack{p \mid \det \mathbf{h} \\ p \text{ odd inert}}} \lambda_{\mathbf{h}}(p) \ \leq \ \prod_{p \ll \log n} \Big(1 + \frac{1}{p}\Big)^{k-1} \ \ll_k \ \big(\log\log(n+3)\big)^{k-1}.$$

Case 3: The remaining odd inert primes. Here Lemma 5.15 gives $\lambda_{\mathbf{h}}(p) = 1 + O_k(p^{-2})$, and $\sum_p p^{-2} < \infty$ closes the argument.

Putting these together, the first two kinds contribute at most $6^{(k-1)r}$, a constant depending only on k and d; the third kind contributes $\ll_k (\log\log(|\det \mathbf{h}| + 3))^{k-1}$, a finite product for each fixed $\mathbf{h}$; and the last kind converges absolutely to a quantity $\ll_k 1$. This completes the proof. □

7.3 The Ford-style average

Theorem 7.3 (Average of the singular series)**.** *Fix* $k \geq 1$*. As* $H \to \infty$*, with the sum running over integer tuples:*

$$\sum_{0 < h_1 < \cdots < h_k < H} \mathfrak{S}_{\mathbf{h}} = \frac{H^k}{k!}\Big(1 + O_{k,d}\Big(\frac{1}{\log\log H}\Big)\Big).$$

The core idea of the proof, following Ford [For16], is to truncate the Euler product at a cutoff $Y = c \log H$, where $0 < c < 1/2$ is fixed. The primes above Y contribute a factor $1 + o(1)$, uniformly in $\mathbf{h}$, while the primes up to Y are replaced by a periodic function Φ whose period Q is at most $H^{2c+o(1)}$, a power of H strictly smaller than H, and whose sum over a full period is computed exactly from Lemma 7.1 via the Chinese remainder theorem.

Proof. Recall from the proof of Corollary 7.2 that $p = 2$ together with the odd primes dividing d, that is, the $r = \omega(2d)$ primes dividing $2d$, satisfy $\delta_{\{0\}}(p) \geq 1/6$ uniformly. We call these the r *exceptional* primes below. Fix $0 < c < 1/2$, and set $Y := c \log H$; for H large enough (depending only on d), Y exceeds all r exceptional primes, so that only odd inert primes contribute to $p > Y$. Set $L := \log H \, (\log\log H)^k$, and for each $p \leq Y$ set $\alpha_p := \lceil \log L / \log p \rceil$, so that:

$$L \leq p^{\alpha_p} < pL.$$

Let $Q := \prod_{p \le Y} p^{\alpha_p}$, and define the Q-periodic function $\Phi(\mathbf{h}) := \prod_{p \le Y} \lambda_{\mathbf{h}}^{(\alpha_p)}(p)$. Throughout, implied constants may depend on k and d. Let:

$$D := \{\mathbf{h} \in [0,H]^k \cap \mathbb{Z}^k : \ h_1, \dots, h_k \text{ pairwise distinct}\}.$$

The singular series $\mathfrak{S}_{\mathbf{h}}$ is only defined for $\mathbf{h} \in D$ (Definition 4.5), and indeed $\det \mathbf{h} = 0$ for every tuple with a repeated coordinate, whereas the finite product $\Phi(\mathbf{h})$ is defined for every $\mathbf{h} \in \mathbb{Z}^k$. We split the proof into five claims; Claims 1 and 4 concern only tuples in D, while Claim 5 is a statement about the full box.

Claim 1: We have, uniformly in $\mathbf{h} \in D$:

$$\prod_{\substack{p > Y \\ \chi_d(p) = -1}} \lambda_{\mathbf{h}}(p) = 1 + O_k\Big(\frac{1}{\log\log H}\Big).$$

Proof of Claim 1: We split by whether $p \mid \det(\mathbf{h})$. For $p \nmid \det(\mathbf{h})$, Lemma 5.15 gives $\lambda_{\mathbf{h}}(p) = 1 + O_k(p^{-2})$, and $\sum_{p>Y} p^{-2} = O(1/Y)$. For $p \mid \det(\mathbf{h})$, Lemma 5.16 gives the upper bound $\lambda_{\mathbf{h}}(p) \le (1+1/p)^{k-1} = 1 + O_k(1/p)$. For the lower bound, observe that every unit lies in S_p (Proposition 5.5), so $\delta_{\mathbf{h}}(p)$ is at least the measure of the set of $a \in \mathbb{Z}_p$ for which every $a + h_i$ is a unit. This set is the complement of at most k residue classes modulo p, so $\delta_{\mathbf{h}}(p) \ge 1 - k/p$. Since $\delta_{\{0\}}(p) \le 1$, we get $\lambda_{\mathbf{h}}(p) \ge 1 - k/p$. Together these give $\lambda_{\mathbf{h}}(p) = 1 + O_k(1/p) = 1 + O_k(1/Y)$, and there are at most $\omega(\det \mathbf{h}) = O_k(\log H / \log\log H)$ such primes, since $\mathbf{h} \in D$ gives $0 < |\det \mathbf{h}| \le H^{\binom{k}{2}}$ and the number of distinct prime factors of an integer $n \ge 1$ is $O(\log n / \log\log n)$. So we obtain:

$$\begin{aligned}\prod_{\substack{p > Y \\ \chi_d(p) = -1}} \lambda_{\mathbf{h}}(p) &= \big(1 + O_k(1/Y)\big) \cdot \big(1 + O_k(1/Y)\big)^{O_k(\log H/\log\log H)} \\ &= 1 + O_k\Big(\frac{\log H}{Y \log\log H}\Big),\end{aligned}$$

uniformly in $\mathbf{h} \in D$. Substituting $Y = c \log H$, the last expression is $1 + O_k(1/\log\log H)$. This proves Claim 1.

Claim 2: We have $Q \le H^{2c + o(1)}$. In particular, setting $\eta := (1-2c)/2 > 0$, we have $Q \le H^{1-\eta}$ for all sufficiently large H.

Proof of Claim 2: Since $p^{\alpha_p} < pL$ for every $p \le Y$, we have:

$$\log Q = \sum_{p \le Y} \alpha_p \log p < \sum_{p \le Y} (\log L + \log p) = \pi(Y) \log L + \vartheta(Y),$$

where $\vartheta(Y) := \sum_{p \le Y} \log p$. By the prime number theorem, $\pi(Y) \sim Y/\log Y$ and $\vartheta(Y) \sim Y$. Moreover $\log L = \log\log H + k \log\log\log H \sim \log\log H \sim \log Y$, so $\pi(Y) \log L \sim Y$. Putting these together we get:

$$\log Q < (2 + o(1))\, Y = (2c + o(1)) \log H,$$

that is, $Q \le H^{2c+o(1)}$. Since $2c < 1 - \eta$, the second statement follows. This proves Claim 2.

Claim 3: We have $\|\Phi\|_\infty \ll_{k,d} (\log\log H)^{k-1}$, and the identical bound holds for $\prod_{p \le Y} \lambda_{\mathbf{h}}(p)$, uniformly in $\mathbf{h} \in D$.

Proof of Claim 3: At an odd split prime we have $S_p = \mathbb{Z}_p$ (Proposition 5.4), so every factor at p, whether truncated or exact, equals 1. We split the remaining primes $p \le Y$ into the r exceptional primes and the odd inert primes $\le Y$. For the exceptional primes, the $0 \le$ part of Lemma 5.14 (respectively Lemma 6.11 at $p = 2$) holds unconditionally, not just for α large, so $\delta_{\{0\}}^{(\alpha_p)}(p) \ge \delta_{\{0\}}(p) \ge 1/6$ directly. The same subset argument as in the proof of Corollary 7.2, applied to the truncated density, gives $\delta_{\mathbf{h}}^{(\alpha)}(p) \le \delta_{\{0\}}^{(\alpha)}(p)$: indeed, "$a + h_i \in S_p \bmod p^\alpha$ for all i" is a stronger condition on a than for a single i, by the identical translation-invariant reasoning as in the untruncated case. Hence $\lambda_{\mathbf{h}}^{(\alpha_p)}(p) \le \delta_{\{0\}}^{(\alpha_p)}(p)^{1-k} \le 6^{k-1}$, so these r primes contribute at most $6^{(k-1)r}$ to Φ, a constant depending only on k, d.

For the odd inert primes $\le Y$, Proposition 5.5 gives $\delta_{\{0\}}(p) = p/(p+1)$ exactly, and the same $0 \le$ part of Lemma 5.14 gives $\delta_{\{0\}}^{(\alpha_p)}(p) \ge p/(p+1)$. So the subset argument gives $\lambda_{\mathbf{h}}^{(\alpha_p)}(p) \le (1+1/p)^{k-1}$, which is identical to the untruncated bound of Lemma 5.16. We find:

$$\prod_{\substack{p\le Y\\ p \text{ odd inert}}} \lambda_{\mathbf{h}}^{(\alpha_p)}(p) \;\le\; \prod_{p\le Y}\Big(1+\frac{1}{p}\Big)^{k-1} \;\ll_k\; \exp\Big((k-1)\sum_{p\le Y}\frac{1}{p}\Big)$$

$$\overset{(*)}{\ll_k}\; (\log Y)^{k-1} \;\ll_k\; (\log\log H)^{k-1},$$

where $(*)$ is Mertens' second theorem, and the last step uses $\log Y \sim \log\log H$. Combining the two kinds of primes we get $\|\Phi\|_\infty \ll_{k,d} (\log\log H)^{k-1}$, and the identical bound holds for $\prod_{p\le Y}\lambda_{\mathbf{h}}(p)$ by the same computation with α_p dropped. This proves Claim 3.

Claim 4: We have $\Big|\prod_{p\le Y}\lambda_{\mathbf{h}}(p) - \Phi(\mathbf{h})\Big| \ll (\log\log H)^{-2}$, uniformly in $\mathbf{h}\in D$.

Proof of Claim 4: For $p \le Y$, write $a := \delta_{\mathbf{h}}(p)$, $a' := \delta_{\mathbf{h}}^{(\alpha_p)}(p)$, $b := \delta_{\{0\}}(p)$, and $b' := \delta_{\{0\}}^{(\alpha_p)}(p)$. By Lemma 5.14 (or Lemma 6.11 at $p=2$), we have $0 \le a'-a \le 4k/p^{\alpha_p}$ and $0 \le b'-b \le 4/p^{\alpha_p}$ (the constant 4 is needed only at $p=2$), and $b, b' \ge 1/6$ throughout, by the proof of Claim 3. Moreover $a' \le b' \le 1$ (the subset argument, trivially), and $|1/b^k - 1/b'^k| \le k\max(1/b,1/b')^{k+1}(b'-b)$ by the mean value bound. This gives:

$$\begin{aligned}\big|\lambda_{\mathbf{h}}(p) - \lambda_{\mathbf{h}}^{(\alpha_p)}(p)\big| &= \Big|\frac{a}{b^k} - \frac{a'}{b'^k}\Big| \\ &\overset{(*)}{\le}\; \frac{a'-a}{b^k} + a'\Big|\frac{1}{b^k}-\frac{1}{b'^k}\Big| \\ &\overset{(**)}{\le}\; \frac{4k\cdot 6^k}{p^{\alpha_p}} + \frac{4k\cdot 6^{k+1}}{p^{\alpha_p}} \;=\; \frac{C_k}{p^{\alpha_p}},\end{aligned}$$

for a constant C_k depending only on k. Here $(*)$ follows from the triangle inequality after adding and subtracting a'/b^k, and $(**)$ from the bounds on $a'-a$, b, b', and a' just recorded, together with the mean value bound.

To pass from this per-prime bound to a bound on the difference of the two full products, write $p_1 < \cdots < p_M$ for the primes $\le Y$ ($M = \pi(Y)$), and for $0 \le j \le M$ define the *hybrid* product with the first j factors truncated and the rest exact:

$$\Pi_j := \lambda_{\mathbf{h}}^{(\alpha_{p_1})}(p_1)\cdots\lambda_{\mathbf{h}}^{(\alpha_{p_j})}(p_j)\cdot\lambda_{\mathbf{h}}(p_{j+1})\cdots\lambda_{\mathbf{h}}(p_M).$$

Then $\Pi_0 = \prod_{p\le Y}\lambda_{\mathbf{h}}(p)$ and $\Pi_M = \Phi(\mathbf{h})$, so telescoping gives:

$$\prod_{p\le Y}\lambda_{\mathbf{h}}(p) - \Phi(\mathbf{h}) \;=\; \Pi_0 - \Pi_M \;=\; \sum_{j=1}^{M}(\Pi_{j-1}-\Pi_j).$$

Since Π_{j-1}, Π_j share every factor except the j-th (exact in Π_{j-1}, truncated in Π_j), we have:

$$\Pi_{j-1} - \Pi_j \;=\; \Big(\prod_{i<j}\lambda_{\mathbf{h}}^{(\alpha_{p_i})}(p_i)\cdot\prod_{i>j}\lambda_{\mathbf{h}}(p_i)\Big)\cdot\big(\lambda_{\mathbf{h}}(p_j) - \lambda_{\mathbf{h}}^{(\alpha_{p_j})}(p_j)\big).$$

To bound the product of the $M-1$ surrounding factors, set $c_p := 6^{k-1}$ if p is one of the r exceptional primes, $c_p := (1+1/p)^{k-1}$ if p is odd inert, and $c_p := 1$ if p is odd split. By the proof of Claim 3, every factor at p, whether truncated or exact, satisfies $0 \le \lambda_{\mathbf{h}}(p) \le c_p$ and $0 \le \lambda_{\mathbf{h}}^{(\alpha_p)}(p) \le c_p$, the lower bound holding because each factor is a ratio of densities. Hence the product of the surrounding factors is at most $\prod_{i\ne j} c_{p_i}$, and since $c_{p_j} \ge 1$ we get:

$$\prod_{i<j}\lambda_{\mathbf{h}}^{(\alpha_{p_i})}(p_i)\cdot\prod_{i>j}\lambda_{\mathbf{h}}(p_i) \;\le\; \prod_{i\ne j} c_{p_i} \;\le\; \prod_{p\le Y} c_p \;\ll_{k,d}\; (\log\log H)^{k-1},$$

where the last bound is the computation in the proof of Claim 3. Combined with the per-prime bound above, $|\Pi_{j-1} - \Pi_j| \ll_{k,d} (\log\log H)^{k-1} \cdot C_k / p_j^{\alpha_{p_j}}$, and summing over $j = 1, \dots, M$ we find:

$$\Big| \prod_{p \le Y} \lambda_{\mathbf{h}}(p) - \Phi(\mathbf{h}) \Big| \le \sum_{j=1}^{M} |\Pi_{j-1} - \Pi_j| \ll_{k,d} (\log\log H)^{k-1} \sum_{p \le Y} p^{-\alpha_p}$$
$$\le (\log\log H)^{k-1} \cdot \frac{\pi(Y)}{L} \ll \frac{(\log\log H)^{k-1}}{(\log\log H)^{k+1}} = \frac{1}{(\log\log H)^2},$$

uniformly in $\mathbf{h} \in D$, using $p^{\alpha_p} \ge L$ and $\pi(Y) \ll \log H / \log\log H$ (by the prime number theorem, as in the proof of Claim 2). This proves Claim 4.

Claim 5: We have $\sum_{\mathbf{h} \in [0,H]^k \cap \mathbb{Z}^k} \Phi(\mathbf{h}) = H^k + O\big(H^{k-\eta} (\log\log H)^{k-1}\big)$.

Proof of Claim 5: Since Φ is Q-periodic and $Q \le H^{1-\eta}$ by Claim 2, we may group $[0,H]^k \cap \mathbb{Z}^k$ by residue class mod Q: each class contains $(H/Q)^k + O((H/Q)^{k-1})$ lattice points. By Lemma 7.1, applied at each $p \le Y$ separately and combined via the Chinese remainder theorem (the moduli p^{α_p}, $p \le Y$, being pairwise coprime), we get:

$$\sum_{\mathbf{a} \bmod Q} \Phi(\mathbf{a}) = \prod_{p \le Y} \sum_{\mathbf{a} \bmod p^{\alpha_p}} \lambda_{\mathbf{a}}^{(\alpha_p)}(p) = \prod_{p \le Y} p^{k\alpha_p} = Q^k.$$

Hence, using Claims 2 and 3:

$$\sum_{\mathbf{h} \in [0,H]^k \cap \mathbb{Z}^k} \Phi(\mathbf{h}) = \Big(\frac{H}{Q}\Big)^k Q^k + O\Big(\Big(\frac{H}{Q}\Big)^{k-1} \|\Phi\|_\infty Q^k\Big)$$
$$= H^k + O\big(H^{k-1} Q (\log\log H)^{k-1}\big) = H^k + O\big(H^{k-\eta} (\log\log H)^{k-1}\big),$$

where we used $Q \le H^{1-\eta}$. This proves Claim 5.

We now put all of our claims to work. The tuples in $[0,H]^k \cap \mathbb{Z}^k$ with a repeated coordinate number $O(H^{k-1})$, and on each of them $0 \le \Phi(\mathbf{h}) \le \|\Phi\|_\infty$, so by Claims 5 and 3:

$$\sum_{\mathbf{h} \in D} \Phi(\mathbf{h}) = H^k + O\big(H^{k-\eta} (\log\log H)^{k-1}\big) + O\big(H^{k-1} (\log\log H)^{k-1}\big) = H^k \big(1 + O(H^{-\eta/2})\big).$$

From now on every tuple lies in D, where $\mathfrak{S}_{\mathbf{h}}$ is defined and Claims 1 and 4 apply. By Claim 4, replacing $\Phi(\mathbf{h})$ by $\prod_{p \le Y} \lambda_{\mathbf{h}}(p)$ changes each of the $O(H^k)$ terms by $O((\log\log H)^{-2})$, so that:

$$\sum_{\mathbf{h} \in D} \prod_{p \le Y} \lambda_{\mathbf{h}}(p) = H^k \Big(1 + O\Big(\frac{1}{(\log\log H)^2}\Big)\Big).$$

Every term of this sum is nonnegative, and multiplying it by the tail factor of Claim 1, which is $1 + O(1/\log\log H)$ uniformly in $\mathbf{h} \in D$, gives $\mathfrak{S}_{\mathbf{h}}$. Hence:

$$\sum_{\mathbf{h} \in D} \mathfrak{S}_{\mathbf{h}} = H^k \Big(1 + O\Big(\frac{1}{\log\log H}\Big)\Big).$$

Finally, each k-subset of $(0,H) \cap \mathbb{Z}$ is counted $k!$ times among the tuples in D, once for each ordering. The remaining tuples in D, those with a coordinate equal to 0 or to H, number $O(H^{k-1})$, and by Claims 1 and 3 each of them has $\mathfrak{S}_{\mathbf{h}} \ll (\log\log H)^{k-1}$. We conclude:

$$\sum_{0 < h_1 < \cdots < h_k < H} \mathfrak{S}_{\mathbf{h}} = \frac{H^k}{k!} \Big(1 + O\Big(\frac{1}{\log\log H}\Big)\Big),$$

where the implied constant depends on k and d. This completes the proof. □

8 Proof of Theorem 1.2

In this section we prove Theorem 1.2 by the method of moments, loosely following Gallagher [Gal76]: under Hypothesis $U(k)$, every factorial moment of the number of elements of E in a short window converges to the corresponding moment of a Poisson distribution.

Fix $\tau > 0$. By §2 (Inputs A and B combined), we have $\rho_1(x) = N(x)/x \sim b(d)/\sqrt{\log x}$. By the hypothesis of Theorem 1.2, the window $y = y(x, \tau)$ satisfies:

$$y \cdot \rho_1(x) \longrightarrow \tau \qquad (\text{so } y \sim \tau\sqrt{\log x}/b(d)),$$

and, for n uniform on the integers $1 \le n \le x$, we let Y_x count $E \cap (n, n+y]$, that is, $Y_x = \sum_{0<t\le y} \mathbf{1}_E(n+t)$. Note that $y \asymp \sqrt{\log x}$, so that $(0, y] \subset (0, C\sqrt{\log x}\,]$ for a suitable constant C, which is a range covered by Hypothesis $U(k)$.

Theorem 8.1 (Factorial moments, conditional on Hypothesis $U(k)$)**.** *For every fixed $k \ge 1$, the k-th factorial moment $\mathbb{E}_n[(Y_x)_k] \to \tau^k$ as $x \to \infty$.*

The idea of the proof is to write the factorial moment as a sum of k-point correlations over k-tuples of shifts in the window, to replace each k-point correlation by the singular series using Hypothesis $U(k)$, and then to evaluate the resulting sum with Theorem 7.3.

Proof. We start by expanding the factorial moment as a sum over ordered k-tuples of distinct shifts in $(0, y]$. Since n is uniform on the $\lfloor x \rfloor$ integers $1 \le n \le x$, while $\rho_k(\mathbf{h}; x)$ is normalized by x, we get:

$$\mathbb{E}_n[(Y_x)_k] = \frac{x}{\lfloor x \rfloor} M_k(x), \qquad \text{where} \quad M_k(x) := k! \sum_{0<h_1<\cdots<h_k\le y} \rho_k(\mathbf{h}; x).$$

Since $x/\lfloor x \rfloor \to 1$, it suffices to show that $M_k(x) \to \tau^k$. By Hypothesis $U(k)$, uniformly over this range, $\rho_k(\mathbf{h}; x) = \mathfrak{S}_{\mathbf{h}} \rho_1(x)^k + o(\rho_1(x)^k)$. The error term is additive, since $\mathfrak{S}_{\mathbf{h}}$ may vanish. There are at most y^k tuples in the range, so summing gives:

$$M_k(x) = k!\, \rho_1(x)^k \sum_{0<h_1<\cdots<h_k\le y} \mathfrak{S}_{\mathbf{h}} \;+\; o\big((y\rho_1(x))^k\big).$$

This range agrees with that of Theorem 7.3 with $H = y$, except possibly for tuples with $h_k = y$. There are $O(y^{k-1})$ such tuples, each with $|\det \mathbf{h}| \le y^{\binom{k}{2}}$ and hence contributing $\mathfrak{S}_{\mathbf{h}} \ll_{k,d} (\log\log y)^{k-1}$ by Corollary 7.2, so that they contribute $O_{k,d}(y^{k-1}(\log\log y)^{k-1}) = o(y^k)$ in total. By Theorem 7.3 with $H = y$, the sum is therefore $y^k(1/k! + o(1))$, so that:

$$M_k(x) = k! \cdot \rho_1(x)^k \cdot y^k\Big(\frac{1}{k!} + o(1)\Big) + o\big((y\rho_1(x))^k\big) = \big(y\rho_1(x)\big)^k(1 + o(1)) \longrightarrow \tau^k,$$

directly from $y\rho_1(x) \to \tau$. This completes the proof. □

Proof (of Theorem 1.2). The factorial moments of Poisson(τ) are exactly τ^k, so by Theorem 8.1 every factorial moment of Y_x, and hence every ordinary moment, converges to that of Poisson(τ). Since the Poisson distribution is determined by its moments, the method of moments [Bil95, §30] gives $Y_x \to$ Poisson(τ) in distribution. Since $Y_x = N(n+y) - N(n)$ takes integer values, this means precisely that for every fixed $m \ge 0$:

$$\mathbb{P}(Y_x = m) \;=\; \frac{1}{\lfloor x \rfloor} \#\big\{1 \le n \le x :\; N(n+y) - N(n) = m\big\} \;\longrightarrow\; e^{-\tau}\frac{\tau^m}{m!}.$$

Since $\lfloor x \rfloor / x \to 1$, the same limit holds with $\lfloor x \rfloor$ replaced by x, which is the statement of Theorem 1.2. This completes the proof. □

Combined with §3, this also completes the proof of Theorem 1.1.

9 Numerical evidence for Hypothesis $U(k)$

In this section we test Hypothesis $U(k)$ numerically for $k = 2, 3$, by comparing the normalized k-point correlations $\rho_k(\mathbf{h}; x)/\rho_1(x)^k$ of E_R (Definition 4.1) with the singular series $\mathfrak{S}_{\mathbf{h}}$ computed from the exact local densities of §5 and §6, and then by looking directly at the window counts. The computations were carried out in the Wolfram Language.

9.1 Method

We took $d \in \{-3, -4, -7, -8, -12, -16, -20\}$, with both genera for $d = -20$ (those of $x^2 + 5y^2$ and $2x^2 + 2xy + 3y^2$); the discriminants -12 and -16 are not fundamental, and $d = -4$ is the case of sums of two squares studied in [FKR17]. For each genus R we sieved the indicator function of $E = E_R$ up to $x = 10^7$ from the values of every reduced form in R, and computed $\rho_k(\mathbf{h}; x)/\rho_1(x)^k$ exactly, taking $h_1 = 0$. This is harmless: $\mathfrak{S}_{\mathbf{h}}$ is invariant under translating $\mathbf{h}$, while translating $\mathbf{h}$ by $-h_1$ only shifts the range of n in Definition 4.1 by $h_1 \le H$, so it changes $\rho_k(\mathbf{h}; x)$ by at most H/x, and hence the quantity in Hypothesis $U(k)$ by $O\big(H/(x\rho_1(x)^k)\big) = O\big(H(\log x)^{O(1)}/x\big)$.

We evaluated $\mathfrak{S}_{\mathbf{h}} = \prod_p \lambda_{\mathbf{h}}(p)$ from the local densities. Odd split primes contribute 1, and for odd inert primes $p \nmid \det(\mathbf{h})$ we used the exact values $\delta_{\mathbf{h}}(p) = 1 - k/(p+1)$ and $\delta_{\{0\}}(p) = p/(p+1)$ (Lemma 5.15 and Proposition 5.5). At the remaining primes, that is $p = 2$, the primes dividing d, and the inert primes dividing $\det(\mathbf{h})$, we computed the truncated densities $\delta_{\mathbf{h}}^{(\alpha)}(p)$ by enumerating residues modulo $p^\alpha \ge 2000$, which by Lemmas 5.14 and 6.11 is accurate to within $O_k(1/2000)$. Primes up to 10^5 were included, so the neglected tail is $O(k^2/(10^5 \log 10^5))$.

For $k = 2$ we took all shifts $(0, h)$ with $1 \le h \le 60$, and for $k = 3$ we sampled 40 random triples $(0, h_2, h_3)$ with $0 < h_2 < h_3 \le 30$. Recall from §4 that a tuple is inadmissible if $\mathfrak{S}_{\mathbf{h}} = 0$, that is, if $\delta_{\mathbf{h}}(p) = 0$ for some prime p, as for $\mathbf{h} = \{0, 1, 2\}$ and $d = -3$; no n then has all of $n + h_i$ in E. Such tuples are excluded from the relative errors below. Hypothesis $U(k)$ concerns shifts up to $C\sqrt{\log x}$, and $\sqrt{\log x}$ is only about 4 at $x = 10^7$, so our shifts up to 60 correspond to rather large values of C.

9.2 2-point and 3-point correlations

Figure 1 shows that the 2-point correlations follow the singular series closely through all of its oscillations. For $d = -3$ (not shown) the largest relative error over $h \le 20$ is 5.9% and the mean relative error over $1 \le h \le 60$ is 3.1%. For $d = -16$ the singular series vanishes at every $h \equiv 2 \pmod 4$, and the empirical 2-point correlation is indeed 0 there. The remaining discrepancies are systematic: the singular series is slightly too high at its peaks and slightly too low in its troughs, as if the empirical 2-point correlation were a little closer to 1. The other discriminants behave in the same way. The data for $d = -12$ coincide with those for $d = -3$, because $x^2 + 3y^2$ and $x^2 + xy + y^2$ represent the same integers; this is still a consistency check, since the local factors at 2 and 3 are computed separately for the two discriminants.

The two genera of discriminant -20 are instructive. They have the same 2-point singular series: at $p = 2$ and $p = 5$ their local value sets are, up to the point 0, complements of each other, each of measure $1/2$, which forces the local factors to agree, and at every other prime the two sets coincide. Their empirical 2-point correlations differ by a few percent at $x = 10^7$ (at $h = 1$ they are 0.817 and 0.757, against $\mathfrak{S}_{\{0,1\}} \approx 0.78$, and at $h = 8$ they are 1.210 and 1.302, against $\mathfrak{S}_{\{0,8\}} \approx 1.27$), but this looks like a lower-order effect: the mean absolute difference over $h \le 20$ decreases from 0.116 at $x = 10^4$ to 0.052 at $x = 10^7$. The genus does affect the singular series for $k = 3$. The values of $x^2 + 5y^2$ are $\equiv 0, \pm 1 \pmod 5$ and those of $2x^2 + 2xy + 3y^2$ are $\equiv 0, \pm 2 \pmod 5$, so $\{0, 1, 2\}$ is inadmissible only for the second genus and $\{0, 1, 3\}$ only for the first. Up to 10^7, the tuple $\{0, 1, 2\}$ occurs 10,561 times for $x^2 + 5y^2$ and never for $2x^2 + 2xy + 3y^2$, while $\{0, 1, 3\}$ occurs 11,207 times for $2x^2 + 2xy + 3y^2$ and never for $x^2 + 5y^2$.

Figure 2 shows that the 3-point correlations lie close to the diagonal over the whole range of values of $\mathfrak{S}_{\mathbf{h}}$, with deviations of a few percent and again slightly below the diagonal at the largest values. The same holds for the other discriminants.

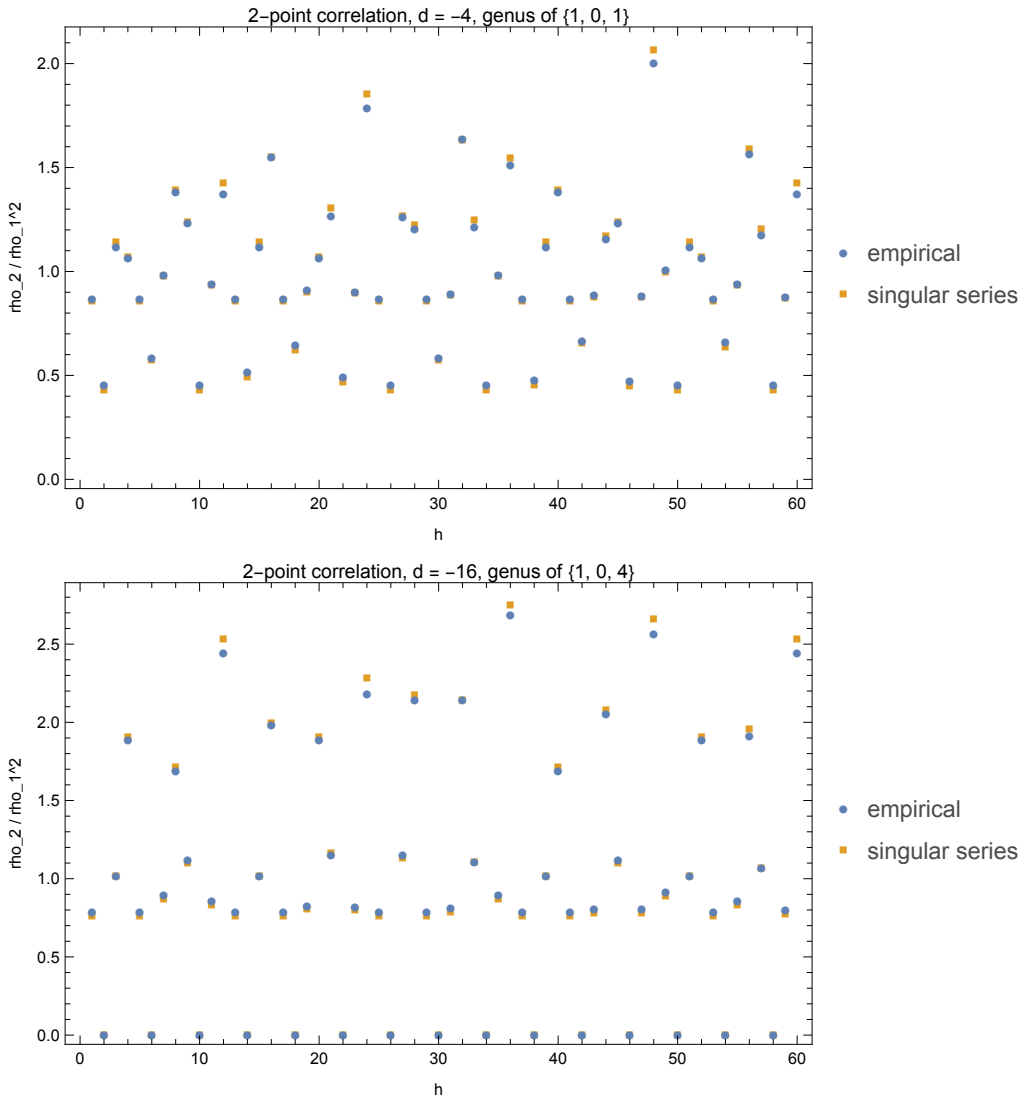


Figure 1: 2-point correlation $\rho_2(\{0,h\};x)/\rho_1(x)^2$ at $x = 10^7$ (blue dots) against the singular series $\mathfrak{S}_{\{0,h\}}$ (orange squares), for $d = -4$ (top) and $d = -16$ (bottom).

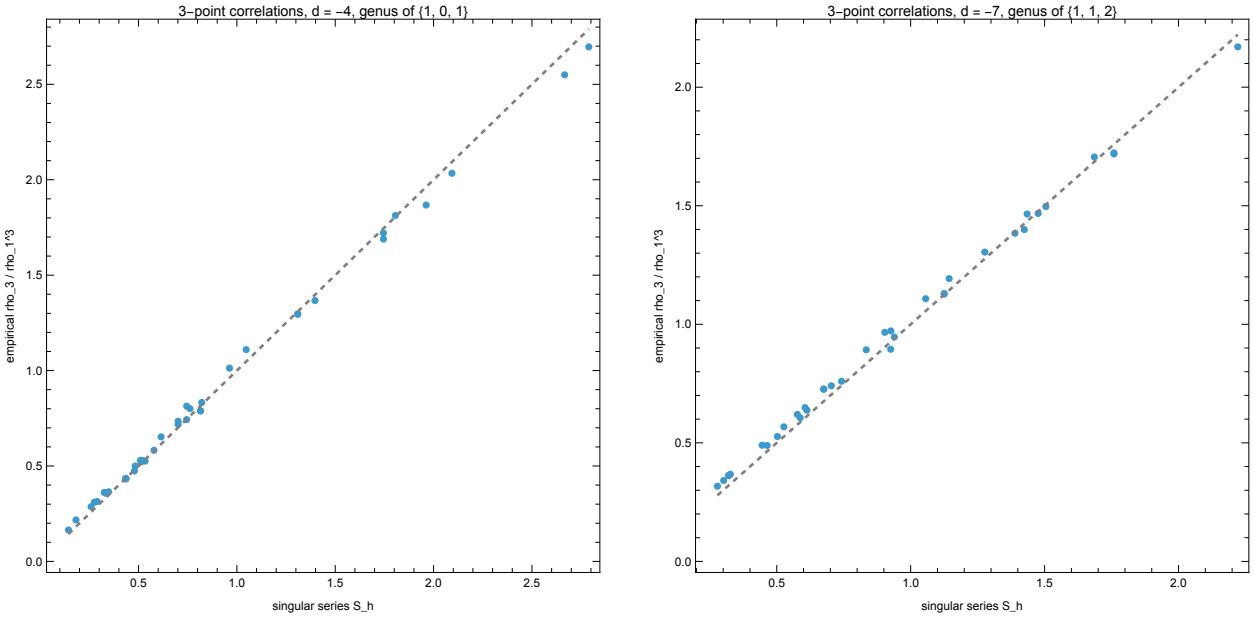


Figure 2: 3-point correlations $\rho_3(\{0,h_2,h_3\};x)/\rho_1(x)^3$ at $x=10^7$ against $\mathfrak{S}_{\{0,h_2,h_3\}}$ for 40 random triples, for $d=-4$ (left) and $d=-7$ (right). The dashed line is the identity.

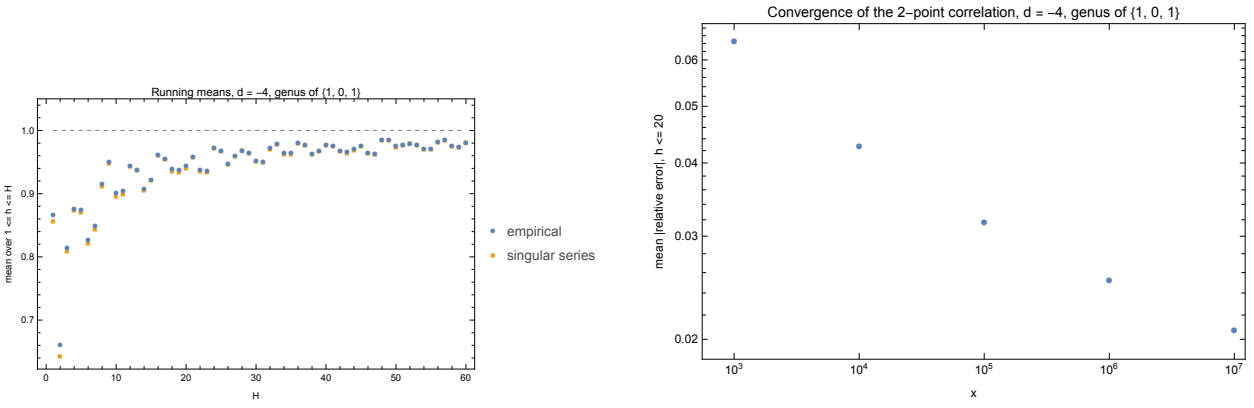


Figure 3: Left: the running means $\frac{1}{H}\sum_{h=1}^{H}\rho_2(\{0,h\};x)/\rho_1(x)^2$ (blue dots) and $\frac{1}{H}\sum_{h=1}^{H}\mathfrak{S}_{\{0,h\}}$ (orange squares) for $1\le H\le 60$, with $d=-4$ and $x=10^7$; the dashed line is the constant 1. Right: the mean relative error $\varepsilon(x)$ of the 2-point correlation, defined in (1), for $d=-4$ and $x=10^3,10^4,10^5,10^6,10^7$, on logarithmic axes.

9.3 Averages and the dependence on x

By Theorem 7.3 with $k = 2$, together with a short monotonicity argument (using $\mathfrak{S}_{\mathbf{h}} \geq 0$), the running means $\frac{1}{H}\sum_{g\leq H}\mathfrak{S}_{\{0,g\}}$ tend to 1. The left panel of Figure 3 shows that the running means of both the singular series and the empirical 2-point correlation do tend to 1, but slowly: for $d = -4$ they are about 0.88 at $H = 5$, 0.90 at $H = 10$, and 0.98 at $H = 60$, and the two sequences are almost indistinguishable.

The right panel shows the dependence on x. Let A be the set of admissible shifts $1 \leq h \leq 20$, that is, those with $\mathfrak{S}_{\{0,h\}} \neq 0$. We measure the agreement at x by the mean relative error:

$$\varepsilon(x) := \frac{1}{|A|}\sum_{h\in A}\left|\frac{\rho_2(\{0,h\};x)}{\rho_1(x)^2\,\mathfrak{S}_{\{0,h\}}} - 1\right|. \tag{1}$$

For $d = -4$ the mean relative error $\varepsilon(x)$ falls from about 6.5% at $x = 10^3$ to about 2% at $x = 10^7$; for $d = -3$ it falls from about 9.5% to about 3%, and for the genus of $2x^2 + 2xy + 3y^2$ at $d = -20$ from about 12% to about 4%. This is much slower than the $x^{-1/2}$ expected from sampling noise alone, which is consistent with the logarithmic rate at which $\rho_1(x) = N(x)/x \sim b(d)/\sqrt{\log x}$ settles down: for $d = -4$ the quantity $N(x)\sqrt{\log x}/x$ is still drifting, from 0.87 at $x = 10^3$ to 0.80 at $x = 10^7$, towards the Landau–Ramanujan constant 0.7642 ....

9.4 The distribution in short windows

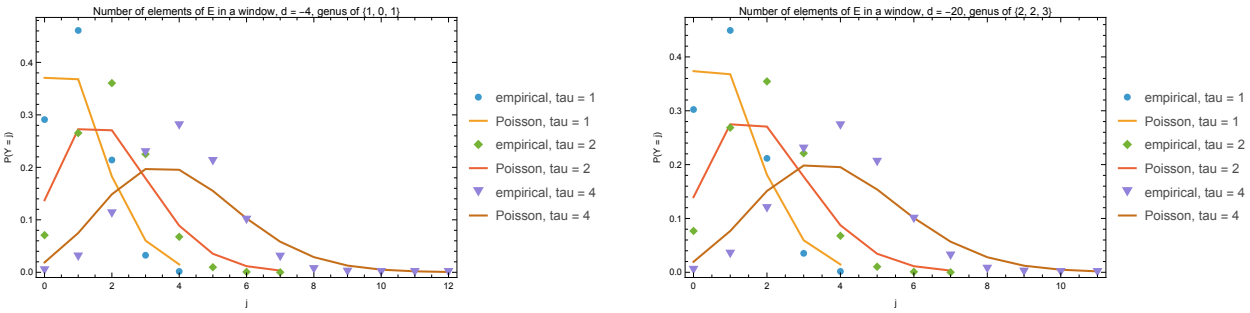


Figure 4: Distribution of the number Y_x of elements of E in a window $(n, n+y]$, $n \leq x = 10^7$, with $y\rho_1(x) \approx \tau$ for $\tau = 1, 2, 4$ (points), against the Poisson distribution with the same mean (lines), for $d = -4$ (left) and for the genus of $2x^2 + 2xy + 3y^2$ at $d = -20$ (right).

Finally, Figure 4 examines the conclusion of Theorem 1.2 directly. At $x = 10^7$ the distribution is clearly not Poisson but under-dispersed: for $\tau = 1$ and $d = -4$ the windows contain no element of E in about 29% of cases instead of $e^{-1} \approx 37\%$, and exactly one element in about 46% instead of 37%; the genus of $2x^2 + 2xy + 3y^2$ at $d = -20$ gives 30% and 45%.

This does not contradict the theorem. The window length $y \sim \tau\sqrt{\log x}/b(d)$ grows only like $\sqrt{\log x}$, and at $x = 10^7$ it is about 5 for $\tau = 1$ and about 20 for $\tau = 4$ when $d = -4$. By the proof of Theorem 8.1, the factorial moments of Y_x are governed by the sums of $\mathfrak{S}_{\mathbf{h}}$ over the actual window, and by Figure 3 such sums are still noticeably below their limiting value at these lengths, especially because the small shifts, where $\mathfrak{S}_{\{0,h\}}$ is far from 1, carry more weight (a window $(n, n+y]$ contains $\lfloor y\rfloor - h$ pairs of integers at distance h). A smaller second factorial moment gives exactly the observed under-dispersion. The convergence to the Poisson distribution is therefore extremely slow, and the evidence for Hypothesis $U(k)$ lies instead in the agreement with the singular series in §9.2–§9.3.

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